

Cerebus Neural Processing System

Instructions for Use



Caution

Device for investigational use in laboratory animals or other tests that do not involve human subjects.

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1 SCOPE OF THIS MANUAL

The Cerebus Neural Processing System (Cerebus system) records and processes neural signals from up to 512 surface or penetrating electrodes.

The system can acquire various biopotential signal types, including electrocorticography, electroencephalography, and electromyography. The system can perform real-time signal processing algorithms on neural signals, including noise cancellation, digital filtering, simultaneous extraction of spike and field potentials, and manual and automatic online spike sorting.

The Cerebus Neural Processing System is not a monitoring system. No physiological alarms are provided. The acquisition and display of biopotential signals are for the interpretation and use of the clinician.

2 SYSTEM DESCRIPTION

The flow of data through the Blackrock Data Acquisitions system begins with a biopotential of interest. Potentials are acquired by an electrode well suited to the signal and location being recorded. Acquired signals are buffered, amplified, and digitized by Blackrock headstages. Finally, data is processed, visualized, and saved by the Central Software Suite utilizing a Blackrock Neural Signal Processor. The system can perform real-time signal processing algorithms including: noise cancellation, digital filtering, simultaneous extraction of spike and field potentials, manual and automatic online spike sorting.

2.1 Cerebus system

Figure 2-1: Analog Cerebus system diagram

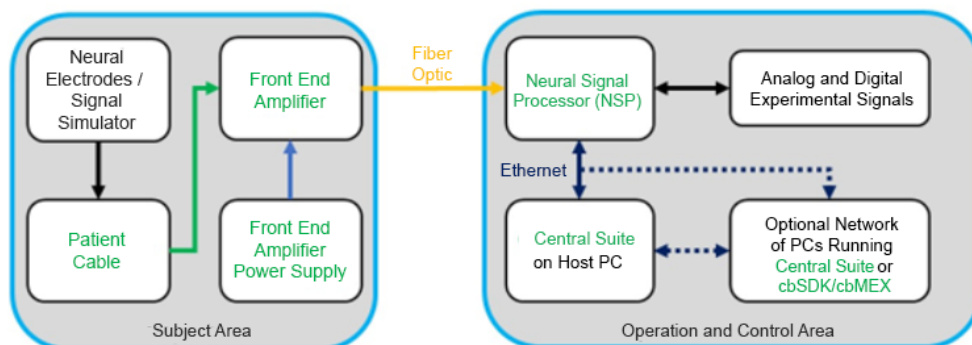
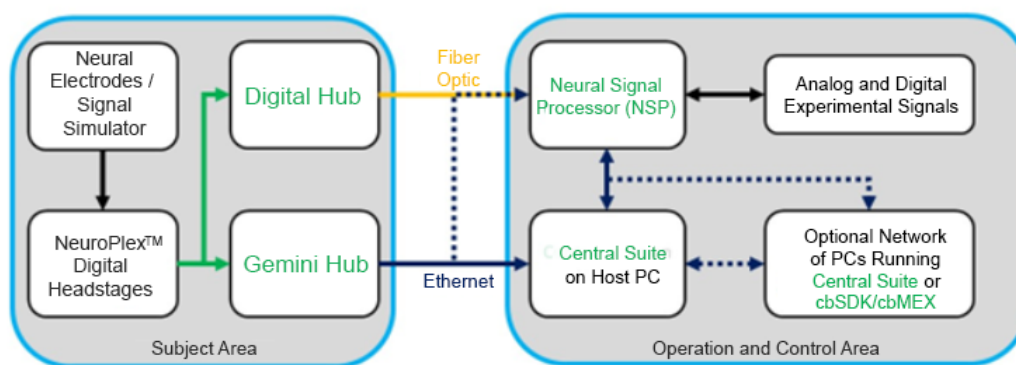


Figure 2-2: Digital Cerebus system diagram



2.2 Intended use

The Cerebus Neural Processing System supports the recording, processing, and display of biopotential signals from user-supplied electrodes. Biopotential signals include electrocorticography (ECoG), electroencephalography (EEG), electromyography (EMG), electrocardiography (ECG), electrooculography (EOG), action potentials (AP), and evoked potentials (EP).

3 CONTRAINDICATIONS, WARNINGS, PRECAUTIONS, AND ADVERSE EFFECTS

3.1 Contraindications



The Cerebus system is a recording system that should not be used in applications involving stimulation.

Contraindications

3.2 Warnings



WARNING

READ THIS ENTIRE MANUAL BEFORE USING THIS DEVICE.



WARNING

A THOROUGH UNDERSTANDING OF THE TECHNICAL PRINCIPLES AND RISKS ASSOCIATED WITH ELECTROPHYSIOLOGICAL RECORDING IS NECESSARY BEFORE USING THIS PRODUCT.



WARNING

COMPLETE THE BLACKROCK MICROSYSTEMS USER TRAINING PROGRAM BEFORE USING THE CEREBUS SYSTEM.



WARNING

ALWAYS OPERATE THE CEREBUS SYSTEM IN A CLEAN ENVIRONMENT.



WARNING

ONLY CONNECT CEREBUS SYSTEM COMPONENTS TO PROPERLY TESTED, GROUNDED, AND DEDICATED AC OUTLETS USING ONLY THE BLACKROCK PROVIDED POWER CABLE TO REDUCE THE RISK OF ELECTRICAL SHOCK OR MALFUNCTION OF PRODUCT. DO NOT USE AN ADAPTER FOR UNGROUNDED WALL OUTLETS.

**WARNING**

DO NOT CONNECT THE CEREBUS SYSTEM TO AN OUTLET CONTROLLED BY A WALL SWITCH, MULTIPLE SOCKET-OUTLET, OR EXTENSION CORD TO AVOID FIRES OR OTHER ELECTRICAL HAZARDS.

**WARNING**

DO NOT TURN ON THE DIGITAL HUB UNTIL CONNECTIONS HAVE BEEN MADE BETWEEN THE SUBJECT AND THE HUB.

**WARNING**

DO NOT USE THE CEREBUS SYSTEM IN THE PRESENCE OF FLAMMABLE ANESTHETIC AGENTS.

**WARNING**

DO NOT USE THE CEREBUS SYSTEM FOR ANY USE OTHER THAN ITS LISTED INTENDED USE.

**WARNING**

AVOID STRONG STATIC DISCHARGES FROM SOURCES LIKE TELEVISIONS OR COMPUTER MONITORS BECAUSE IT CAN DAMAGE THE ELECTRICAL COMPONENTS OF THE SYSTEM.

**WARNING**

KEEP THE CEREBUS SYSTEM AWAY FROM LIQUIDS. CONTACT WITH WATER, SHOWER SPRAY, OR WET SURFACES CAN LEAD TO THE PATIENT RECEIVING AN ELECTRICAL SHOCK.

**WARNING**

CONNECTION OF EXTERNAL INSTRUMENTS MAY COMPROMISE ELECTRICAL SAFETY COMPLIANCE WITH IEC 60601-1.

**WARNING**

THE CEREBUS SYSTEM SHOULD BE DISCONNECTED FROM ANY ELECTRODES DURING CARDIAC DEFIBRILLATION AND OTHER LIFE-SUPPORTING DEVICES.

**WARNING**

THE CONDUCTIVE PARTS OF ELECTRODES AND THEIR CONNECTORS, INCLUDING NEURAL ELECTRODES, SHOULD NOT CONTACT OTHER CONDUCTIVE PARTS INCLUDING EARTH.

**WARNING**

PLACE THE CEREBUS SYSTEM IN A SECURE LOCATION.

**WARNING**

AVOID TRIPPING ON CORDS CONNECTED TO THE CEREBUS SYSTEM.

**WARNING**

DO NOT REPAIR OR PERFORM MAINTENANCE ON THE EQUIPMENT WHILE IT IS BEING OPERATED.

**WARNING**

ONLY CONNECT BLACKROCK-APPROVED EQUIPMENT TO THE CEREBUS SYSTEM.

**WARNING**

DO NOT CONNECT THE COMPUTER HOSTING THE CENTRAL SOFTWARE TO THE INTERNET.

**WARNING**

DISABLING FIREWALLS AND ANTIVIRUS SOFTWARE COULD LEAVE YOUR SYSTEM VULNERABLE TO CYBERATTACKS.

**WARNING**

DO NOT TOUCH ANY EXPOSED ELECTRICAL CONDUCTORS OR USE DAMAGED DATA CABLES WHEN USING THE HUB.

**WARNING**

ONLY CONNECT HEADSTAGES TO THE HUB.

**WARNING**

TO AVOID THE RISK OF ELECTRIC SHOCK, THIS EQUIPMENT MUST ONLY BE CONNECTED TO A SUPPLY MAIN WITH PROTECTIVE EARTH.

3.3 Precautions

**Caution**

Follow the restrictions of use for third-party electrodes or arrays.

**Caution**

Electrically isolate all third-party recording or control systems connected to the Cerebus system and components for patient safety.

**Caution**

Inspect the Cerebus system and cables before use. Do not use damaged cables.

**Caution**

Always completely shut down the hub before disconnecting, reconnecting, or switching headstages. Disconnecting and reconnecting a headstage with the hub powered on can damage the components.

**Caution**

Fiber-optic cable is easily damaged. Do not bend it (bend radius of 5.0 cm) or crush it.

3.4 Adverse effects

There are no adverse effects associated with the Cerebus Neural Processing System.

4 SYMBOLS

Note: Symbols vary based on the component. See the individual component's IFU for specific symbols.



The **Manufacturer** symbol indicates the medical device manufacturer. **Reference Number:** ISO 15223-1:2021 5.1.1



The **Date of Manufacture** symbol indicates the date the system was manufactured and is accompanied by a date. **Reference Number:** ISO 15223-1:2021 5.1.3



The **Use-by Date** symbol indicates the date after which the medical device is not to be used. **Reference Number:** ISO 15223-1:2021 5.1.4



The **Batch Code** symbol indicates the manufacturer's code for identifying the batch or lot. **Reference Number:** ISO 15223-1:2021 5.1.5



The **Catalog Number** symbol indicates the manufacturer's catalog number so that the device may be identified. For Blackrock Microsystems it is called the Part Number (PN). **Reference Number:** ISO 15223-1:2021 5.1.6



The **Serial Number** symbol indicates the manufacturer's serial number so that a specific medical device can be identified. **Reference Number:** ISO 15223-1:2021 5.1.7



The **Sterilized Using Ethylene Oxide** symbol indicates that the device has been sterilized using ethylene oxide. **Reference Number:** ISO 15223-1:2021 5.2.3



The **Do Not Resterilize** symbol indicates that the device should not be resterilized after it has been sterilized. **Reference Number:** ISO 15223-1:2021 5.2.6



The **Non Sterile** symbol indicates that this device has not been subjected to a sterilization process. **Reference Number:** ISO 15223-1:2021 5.2.7



The **Do Not Use if Package is Damaged** symbol indicates that a medical device should not be used if the package has been damaged or opened. **Reference Number:** ISO 15223-1:2021 5.2.8



The **Do Not Reuse** symbol indicates a medical device that is intended for one use, or for use on a single patient during a single procedure. **Reference Number:** ISO 15223-1:2021 5.4.2



The **Consult Instructions for Use** symbol indicates the need for the user to consult the instructions for use (this document). **Reference Number:** ISO 15223-1:2021 5.4.3



The **Caution** symbol indicates the need for the user to consult the instructions for use (this document) for important cautionary information such as warnings and precautions that cannot, for a variety of reasons, be presented on the medical device itself. **Reference Number:** ISO 15223-1:2021 5.4.4, ISO 7000/IEC 60417-0434A 2004-01-15



The **On (Power)** symbol indicates that the system is connected to the mains power, and the system should be considered powered. **Reference Number:** IEC 60417:2002 5007



The **Off (Power)** symbol indicates that the system is disconnected from the mains power. The system may still be powered. **Reference Number:** IEC 60417:2002 5008



The **Standby (Power)** symbol identifies the on / off (power) switch location, and that the system is in standby / low power consumption mode. The system is powered. **Reference Number:** IEC 60417:2002 5009



The **Protective Ground** symbol identifies any terminal which is intended for connection to an external conductor for protection against electric shock in case of a fault, or the terminal of a protective earth (ground) electrode. **Reference Number:** IEC 60417-5134 2003-04-04 5019



The **Dangerous Voltage** symbol indicates hazards arising from dangerous voltages. **Reference Number:** IEC 60417-5036 20020-10-07 5036



The **Electrostatic Sensitive Devices** symbol indicates packages containing electrostatic sensitive devices, or to identify a device or a connector that has not been tested for immunity to electrostatic discharge. **Reference Number:** IEC 60417-5134 2003-04-04 5134



The **Type BF Applied Part** symbol identifies a type BF applied part complying with IEC 60601-1. **Reference Number:** BS EN ISO 60411:2002 IEC 60417-5333



The **MR Unsafe** symbol indicates that the product poses unacceptable risks to the patient, medical staff, or other persons in the MR environment. **Reference Number:** ASTM F2503-20

5 SYSTEM REQUIREMENTS

The following list shows the minimum system requirements for the software.

- Microsoft Windows 10 (x64)
- AMD or Intel 2.0 GHz Dual Core CPU
- 4 GB of RAM
- 1-Gbps or faster Ethernet interface card
- SATA II HDD - 1 TB / 3 Gbps

Improved system specifications might be required depending on the bandwidth of the data streams to be recorded.

Blackrock supplies an optional host computer that is configured and tested by our engineers before it ships with your Cerebus system. Contact sales@blackrockneuro.com for more information.

5.1 *Cybersecurity requirements*

- Do not connect the computer hosting the Central software to the internet.
- Ensure no malware and antivirus background processes are running.
- Ensure that the Windows firewall is not blocking access to IP addresses used by the Cerebus system.
- Require a Windows system password for the host computer.

6 SPECIFICATIONS

- **System Name:** Cerebus Neural Processing System (Cerebus system)
- **Power Requirements:** 120-240 V AC, 50/60 Hz, 6.0 A
- **Line Noise Serviceable Fuses:** 5 x 20 mm, 250 V, 1.6 A, Slow Blow
- **Compliance Standards:** IEC 60601-1, IEC 60601-1-2, TUV listed
- **Type of Protection Against Electric Shock:** Class I
- **Medical Equipment Classification:** Class I ME
- **Degree of Protection:** Type BF Applied Part
- **Mode of Operation:** Continuous
- **Ingress Protection:** Ordinary Equipment, not fluid resistant, IP20.
- **Operating Environment:** 10°C to 35°C, 5 to 85% relative humidity (non-condensing)
- **Storage Environment:** -20°C to 50°C, 5 to 95% relative humidity (non-condensing)
- **Atmospheric Pressure:** > 91 kPa (< 3000 m above sea level)
- **Neural Signal Processor (legacy / fiber optic Neural Signal Processors or Gemini NSP)**
 - **Dimensions:** 17" x 20 $\frac{7}{8}$ " x 3 $\frac{1}{2}$ "
 - **Weight:** 13 $\frac{1}{2}$ lb
- **Digital Hub**
 - **Dimensions:** 5" x 4 $\frac{1}{2}$ " x 1 $\frac{1}{4}$ "
 - **Weight:** 1 lb
- **Gemini Hub**
 - **Dimensions:** 7 $\frac{1}{2}$ " x 6 $\frac{5}{8}$ " x 1 $\frac{3}{8}$ "
 - **Weight:** 2 $\frac{1}{4}$ lb
- **Front End Amplifier:**
 - **Dimensions:** 7 $\frac{3}{4}$ " x 4 $\frac{1}{2}$ " x 2 $\frac{3}{8}$ "
 - **Weight:** 2 $\frac{3}{4}$ lb

Table 1: Guidance and manufacturer's declaration electromagnetic immunity

The Cerebus system is intended for use in the electromagnetic environment specified below. The customer or the user of the Cerebus system should ensure that it is used in such an environment.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment - Guidance
Conducted RF	3 Vrms	3 Vrms	Portable and mobile RF communications equipment should be used no closer to any part of the Cerebus system, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance $d = \sqrt[1.2]{P}$ 150 kHz to 80 MHz $d = \sqrt[1.2]{P}$ 80 MHz to 800 MHz¹ $d = \sqrt[1.2]{P}$ 800 MHz to 2.5 GHz where <i>P</i> is the maximum output power² rating of the transmitter in watts (W) according to the transmitter manufacturer and <i>d</i>³ is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey should be less than the compliance level in each frequency range⁴. Interference may occur in the vicinity of equipment marked with the following symbol: 
IEC 61000-4-6	150 kHz to 80 MHz		
Radiated RF	3 V/m	3 V/m	
IEC 61000-4-3	80 MHz to 2.5 GHz		

Notes:

- 1. At 80 MHz and 800 MHz, the higher frequency range applies.
- 2. These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment - Guidance
3. Field strength from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the Cerebus system is used exceeds the applicable RF compliance level above, the Cerebus system should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the Cerebus system. 4. Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.			

Table 2: Recommended separation distances between portable and mobile RF communications equipment and the Cerebus system

The Cerebus system is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the Cerebus system can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the Cerebus system as recommended below, according to the maximum output power of the communications equipment.

Rated maximum output power of transmitter	Separation distance according to frequency of transmitter	
	150 kHz to 800 MHz $d = \sqrt[1.2]{P}$	800 MHz to 2.5 GHz $d = \sqrt[2.3]{P}$
0.01	0.12	0.23
0.1	0.38	0.73
1	1.2	2.3
10	3.8	7.3
100	12	23

For transmitters rated at a maximum output power not listed above, the recommended separation distance d in meters (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in Watts (W) according to the transmitter manufacturer.

Notes

- At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.
- These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.

7 HARDWARE

The Cerebus system contains the following subcomponents.

Table 1: Cerebus system subcomponents

Component Name	Analog Cerebus system	Digital Analog Cerebus system (Fiber-optic)	Digital Analog Cerebus system (Ethernet)
Analog Front End Amplifier	Required	n/a	n/a
Neural Signal Processor	Required	Required	Optional
Digital Hub	n/a	Required	n/a
Gemini Hub	n/a	n/a	Required

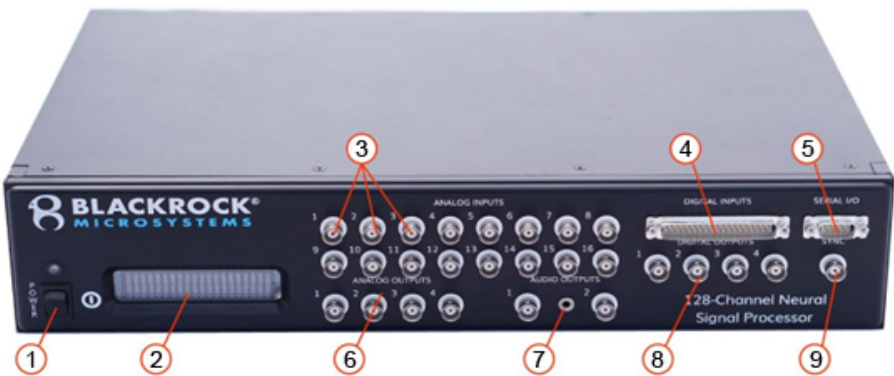
7.1 Neural Signal Processor (NSP)

The NSP is the real-time processor of the system: it digitally processes all signals and data, including providing digital filtering, spike extraction, and spike sorting, and then transmits it to the host computer by using the Ethernet UDP protocol. The NSP has multiple analog and digital inputs and outputs that can be programmed by using the software or one of the supplied Software Development Kits (SDKs). Multiple NSPs may be synchronized to record signals from multiple electrodes.

Note: The fiber optic link input connector has been relocated to the back panel in NSP versions 1.75 and above. The location of this port was previously under the sync port on the front panel.

Figure 7-1: NSP Front

- 1. Power switch
- 2. LCD display
- 3. Analog inputs
- 4. Digital inputs
- 5. Serial I/O
- 6. Analog outputs
- 7. Audio outputs
- 8. Digital outputs
- 9. Sync port.



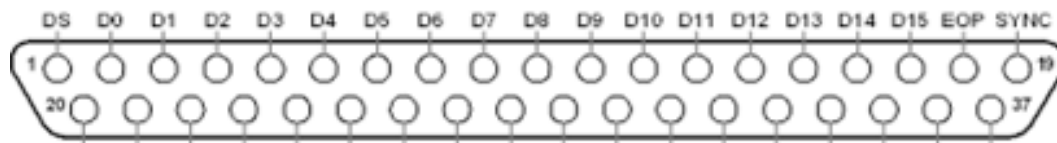
1. **Power Switch** - Use this switch to turn the NSP on and off. The LED above the switch illuminates blue when the unit is on.

Note: On the Gemini NSP and legacy / fiber optic Neural Signal Processors, this switch is a momentary switch to power down the NSP and does not turn main power to the device ON/OFF. See rear panel for mains switch for mentioned part numbers.

On the Gemini NSP, this momentary switch must be held down for 8 seconds to power down the NSP.

2. **LCD Display** - This display shows the current operating status of the unit. The statuses include **Initializing**, **NSP Startup**, **NSP Running**, **NSP Standby**, and **Synchronized**.
3. **Analog Inputs** - Auxiliary analog signals can be recorded through 16 analog input BNC ports. The analog source may range ± 5.0 V and should come from a source impedance of less than 100 Ω . The coupling of each input channel can be manually selected in the software. By default, channels 1-8 are AC-coupled and channels 9-16 are DC-coupled.
4. **Digital Input** - Digital events can be recorded through the 16-bit DB37 input port. The pin diagram is shown below. DS is the digital strobe pin. D0-15 are data pins. EOP is reserved. SYNC is an output pin and can be used with external equipment to indicate when the port is scanned. Input range is 0V-5V TTL levels. The port is polled every 1/30000 of a second. Strobed data is buffered up to 5 strobes per 1/30000 of a second and is latched on the rising edge of the DS pin.

Figure 7-2: Digital in Pin Diagram



Note: All lower pins are connected to the digital ground.

5. **Serial I/O** - The port is an RS232 DB9 digital input/output port. The pin diagram is shown below. Currently, the software only supports this port as an input. Pin 2 is "Receive Data", pin 3 is "Transmit Data", and pin 5 is "Ground". The configuration of the port is: Baud rate: 115200, Data bits: 8, Parity: none, Stop bits: 1, Flow control: disabled.

Figure 7-3: Serial I/O Pin Diagram

Currently, the software only supports this port as an input.

- Pin 2 is "Receive Data"
- Pin 3 is "Transmit Data"
- Pin 5 is "Ground".



The configuration of the port is:

-
- Baud rate: 115200
 - Data bits: 8
 - Parity: none
 - Stop bits: 1
 - Flow control: disabled.
6. **Analog Outputs** - Analog output BNC connectors can be used to send monitoring signals or stimulus waveforms to other connectors.
 - Rated Voltage: ± 5 V
 - Rated Current: ± 20 mA
 - Rated Power: 100 mW
 - Maximum Frequency: 15 kHz
 7. **Audio Output** - The system sends a line-level audio signal of the selected data channel to two BNC ports (Left and Right channeled respectively) and one 3.5mm female stereo audio connector simultaneously.
 - Rated Voltage: ± 1 V
 - Rated Current: ± 20 mA
 - Rated Power: 20 mW
 - Maximum Frequency: 15 kHz
 8. **Digital Outputs** - Four single-bit digital BNC outputs can be programmed for monitoring or timing functions. These ports can be setup to send a TTL signal if spike activity is detected on any neural channel. They can also be configured to output a digital pulse train at a user-defined frequency and duty cycle. Digital Output 1 can also be used for syncing external equipment by sending a unique pulse every 14 seconds. The entire sync pattern will repeat every hour. See the Central Software Suite User Manual for more details.
 - Rated Voltage: +5 V
 - Rated Current: +20 mA
 - Rated Power: 100 mW
 - Maximum Frequency: 15 kHz
 9. **Sync Port** - A synchronization pulse can be set as an optional line to inform external equipment when the NSP neural signal inputs and front panel ports are scanned. It is active on the rising edge of the signal.
 - Rated Voltage: +5 V
 - Rated Current: +20 mA
 - Rated Power: 100 mW
 - Maximum Frequency: 15 kHz

Figure 7-4: NSP Back

Neural Signal Processor



Neural Signal Processor I



Neural Signal Processor 1.75



Neural Signal Processor II



Gemini NSP

10. **Line Noise Cancellation Port** - On the Neural Signal Processor (obsolete), the hardware Line Noise Cancellation receptacle is combined with the main power receptacle. On legacy/fiber optic Neural Signal Processors, a separate power receptacle is used for hardware line noise cancellation. To use this feature, plug a standard power cable into this port and enable the feature in software as described in the Central Suite User Manual. See **To configure the hardware settings for one or more analog inputs:** on page 37.
Note: This receptacle is optional and is not required to power and operate the NSP under normal conditions
11. **Synchronization Port** - This DB9 port is located on the back of the NSP, and it is used to synchronize two NSPs. The synchronization occurs automatically as Central runs on both computers if the sync cable (Blackrock PN-5584) is properly connected between two NSPs. Some models may not have this port. To add synchronization capability to your NSP, contact Blackrock Microsystems support at sales@blackrockneuro.com.
12. **Fiber-optic Link** - This port connects to the Amplifier using a fiber-optic cable. An LED to the right of the connector turns green when a link is established and turns yellow when the link is broken.
13. **Mains Power Entry** - This supplies main power to the NSP and is required to power and operate the NSP.
14. **Mains Power Switch** - Use the Power switch to turn the main power to the NSP on or off. Ensure this switch is in the on position marked by the **I** on the switch.
15. **Network Link** - On Gemini NSPs, the four Ethernet jacks on the back of the unit allow for connection between NSP and host computer, and NSP and Gemini Hub.

7.2 Digital Hub

The Digital Hub receives digital signals from attached digital headstage accessory devices.

Figure 7-5: Digital Hub



The Digital Hub allows multiple digital headstages to be connected. It provides the sampling clock signal to the digital headstages. The hub also converts incoming digital information from the headstages into the optical domain to transfer to the NSP through a fiber-optic cable. This optical data is immune to electromagnetic field interference.

Figure 7-6: Digital Hub front panel

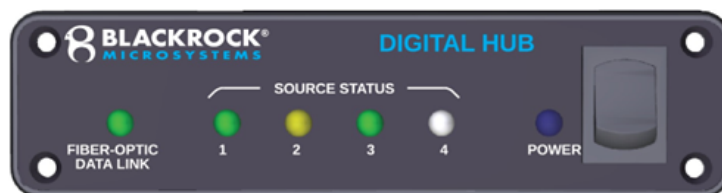
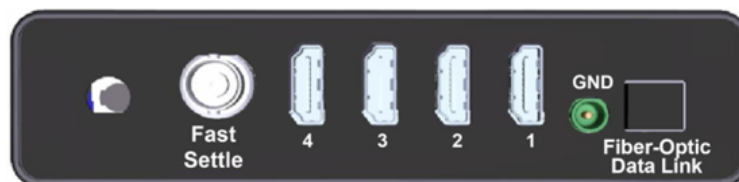


Figure 7-7: Digital Hub back panel



7.2.1 Front panel status LEDs

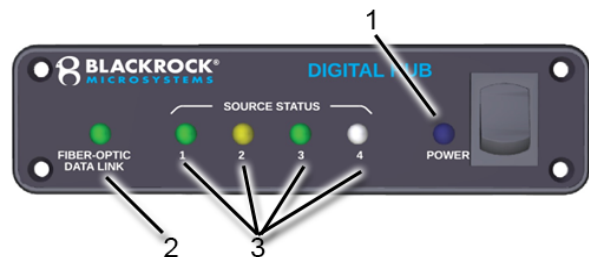
There are six color-coded device status LEDs on the Digital Hub.

Figure 7-8: Digital Hub front panel with status LEDs

1. POWER LED

- **Blue** - The Digital Hub is powered (plugged in).

Note: The Digital Hub is powered by a medical grade power supply that is included with the Digital Hub system.



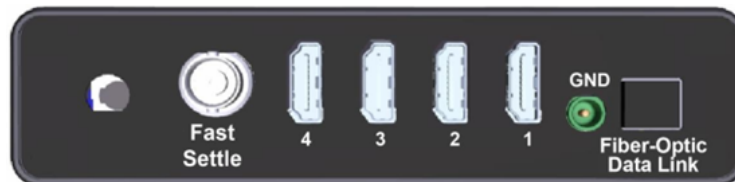
2. FIBER-OPTIC DATA LINK

- **Green** - The NSP and the Digital Hub.
- **Yellow** - No connection.

3. SOURCE STATUS (LEDs 1-4) - These LEDs indicate the status of incoming data streams.

- **Green** - The Digital Hub is receiving data from a connected digital headstage correctly.
- **Yellow** - Data is not being transmitted to the Digital Hub correctly.
- **Not illuminated** - No data is being received or the digital data cable is not plugged into the input slot.

Figure 7-9: Digital Hub back panel



7.2.2 Digital inputs

There are 4 digital inputs (labeled 1-4) on the Digital Hub. The Digital Hub prioritizes HDMI inputs in order from 1 to 4. This means that if two 96-channel Digital Headstages are connected in inputs 2 and 3 respectively, input 2 will send the entire 96 channels to the NSP but input 3 will send only the first 32 channels. The Digital Hub will always send the first 128 channels to the NSP in this way unless there are fewer than 128 channels, in which case it will send all channels.

7.2.3 Patient ground

The green **GND** connector is located on the back panel of the Digital Hub. It is the patient ground connector, which can be used for a reference ground if needed.

7.2.4 Fiber-optic link

The fiber-optic link connector is located on the back panel of the Digital Hub, where it provides a connection to the Blackrock NSP through a fiber-optic cable.

7.2.5 Fast Settle

The Fast Settle input connector is located on the back panel of the Digital Hub. This connector is tied to each of the four Digital Data Cable inputs. The signal fed into the BNC connector passes through the system to the connected digital headstages, providing a fast settle signal to the attached digital headstage(s). When the fast settle input receives a TTL high it will blank the signal, causing the signal to flatline.

7.3 Gemini Hub

The Gemini Hub receives digital signals from attached digital headstage accessory devices.

Figure 7-10: Gemini Hub



Figure 7-11: Gemini Hub front panel

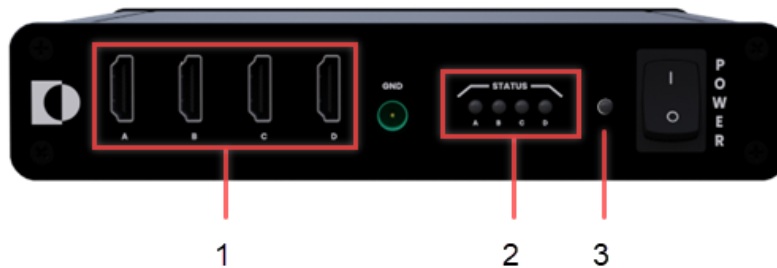


Figure 7-12: Gemini Hub back panel



7.3.1 Front panel ports and status LEDs

Figure 7-13: Front panel LEDs on the Gemini Hub front panel



1. **Digital inputs** - Four ports labeled A-D.
2. **Status LEDs** - The hub provides a color-coded indicator (A-D) for each digital input:
 - **Off / Unlit** - No data is being received or the Digital Data Cable is unplugged from the input slot.
 - **Blinking amber** - No network connection between the NSP and the Gemini Hub.
 - **Steady amber** - Data is not being transmitted to the Gemini Hub correctly.
 - **Steady green** - The hub is correctly receiving data from a connected digital headstage.
3. **LED power indicator**
 - **Off / Unlit** - The Gemini Hub is not powered (unplugged).
 - Gemini Hubs released before October 2023:
 - **Blinking amber** - There is no network connection between the NSP and the Gemini Hub.
 - **Blue** - The Gemini Hub is powered (plugged in).
 - Gemini Hubs released after October 2023:
 - **Blue** - The Gemini Hub is powered (plugged in). If this LED remains blue, there is no network connection between the NSP and the Gemini Hub.
 - **Green** - The connection between the host computer and the Gemini Hub is active.
 - **Green / Blue cycle** - The connection between the host computer and the Gemini Hub is absent or unstable.

Note: Contact Support for the manufacture date of your device.

7.3.2 Digital inputs

Figure 7-14: Digital inputs on the Gemini Hub front panel



There are 4 digital inputs (labeled A-D) on the Gemini Hub. The Gemini Hub prioritizes HDMI inputs in order from 1 (A) to 4 (D). This means that if two 96-channel Digital Headstages are connected in inputs 2 and 3 respectively, input 2 will send the entire 96 channels to the NSP but input 3 will send only the first 32 channels. The Gemini Hub will always send the first 128 channels to the NSP in this way unless there are fewer than 128 channels, in which case it will send all channels.

7.3.3 Patient ground

Figure 7-15: Patient Ground on the Gemini Hub front panel



The green **GND** connector is located on the back panel of the Gemini Hub. It is the patient ground connector, which can be used for a reference ground if needed.

7.3.4 Network link

The network link connector is located on the back panel of the Gemini Hub, where it provides a connection to the NSP through an Ethernet crossover cable.

7.3.5 Fast Settle

The Fast Settle Input connector is located on the back panel of the Gemini Hub. This connector is tied to each of the four Digital Data Cable inputs. The signal fed into the BNC connector is passed through the system to the connected digital headstages, providing a fast settle signal to the attached digital headstages. When the fast settle input receives a TTL high it will blank the signal, causing the signal to flatline.

7.4 Neural Signal Front-End Amplifier

The Neural Signal Front-End Amplifier receives analog signals directly from the electrodes or via headstages (for example, unity-gain voltage followers) depending on the impedance of the electrodes.

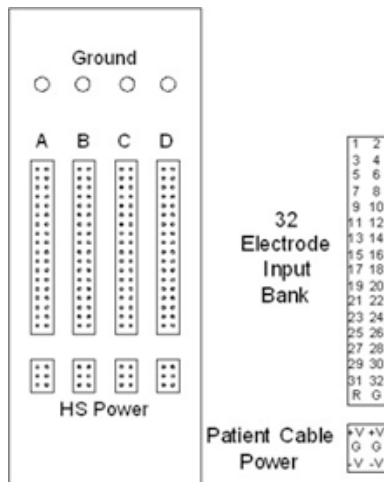
The analog signals are amplified, filtered (first-order high-pass at 0.3 Hz and third-order low-pass at 7,500 Hz), and digitized (30 kHz, 16 bits at 250 nV resolution), converted into the optical domain and then transmitted to the NSP via a fiber-optic link, which is immune to electromagnetic field interference. At a later stage, digital filtering allows these two signals to be separated and recorded in different data files.

Figure 7-16: Neural Signal Front-End Amplifier



The Front-End Amplifier case is not connected to the earth ground. The Front-End Amplifier ground is isolated and floating. The ESD shunt circuits conduct differences of 1000 V or more.

The Front-End Amplifier has four 34-pin banks. Each bank consists of 32 channels, a bank reference pin, and a ground pin. The electrodes in each bank are differentially amplified with respect to the reference input of the same bank. If channels on all banks are to be electrically measured with respect to a common reference, this reference needs to be connected to the reference input on every bank.

Figure 7-17: Neural Signal Front-End Amplifier Banks

Additionally, four 6-pin APS banks provide power (± 5 V) to the headstage and can deliver 130 mA of combined current from the Amplifier. The ground pins on every bank are tied together. Ensure the Front-End Amplifier is the central grounding point for electrical ground connections to the patient.

7.5 Front-End Amplifier power supply (APS)

The Front-End Amplifier power supply (APS) includes five analog and digital supply channels with monitoring, sequencing, and emergency shutdown controls.

Figure 7-18: Front-End Amplifier Power Supply

To turn on the APS:

1. Press the **On / Off** switch on the back of the unit. The APS will start a power-up sequence of analog and digital supplies.
2. Ensure the green **ON LED** illuminates to confirm the power-up sequence is complete.

Note: When the power switch is turned off, this sequence will be reversed and the output of the APS will be tied to ground.

To reset the APS following a voltage or power delivery error:

1. Ensure the APS has shut down and a red error LED is illuminated.
2. Check the Front-End Amplifier and the patient cable connections for any inadvertent shorting.
3. Turn off the APS and then turn it on again.

7.6 Cables

The system includes the following cables.

Note: Standard power cables for the NSP and Hub are included with the system but are not shown.

Fiber-optic cable

Use this cable to connect the Digital Hub to the Neural Signal Processor.

Figure 7-19: Fiber-optic Cable



Crossover cable

Use Ethernet crossover cables to make the following connections:

- The first cable connects the Gemini Hub to the Neural Signal Processor (NSP), and the second connects the NSP to the host computer.
- One cable connects the Gemini Hub to the host computer.

Figure 7-20: Crossover Cable



Front-end Amplifier Power Supply cable

Use this cable to connect the Front-end Amplifier to its power supply.

Figure 7-21: Front-end Amplifier Power Supply Cable



8 ACCESSORIES

Blackrock sells system accessories, including headstages for interfacing the patient-contacting electrodes to the system.

Digital headstages can be attached to the Hub.

8.1 Digital headstages

Digital headstages are required to allow the Digital/Gemini Cerebus system to interface with electrodes. Select a digital headstage based on the connector type of the electrodes.

Note: These connector types and the selection process are not explained here but are explained headstage IFUs.

The following digital headstage is currently available:

- CerePlex family of headstages

8.2 Analog headstages

The high-input impedance and low-bias current of the Amplifier inputs enable microelectrodes with 20 k Ω (at 1kHz) or lower impedance values to be connected directly to the Amplifier inputs without requiring a headstage. This configuration has the advantage of avoiding noise added to the signals by the headstage, but makes the application more susceptible to environmental electromagnetic noise.

To minimize environmental noise, keep direct electrode connections shorter than 20 cm (8 inches). If longer connections are needed, use Blackrock headstages. Headstages allow microelectrodes with impedances up to 5 M Ω (at 1 KHz).

Select an analog headstage based on the connector type on the electrodes.

Note: These connector types and the selection process are not explained here but are explained headstage IFUs.

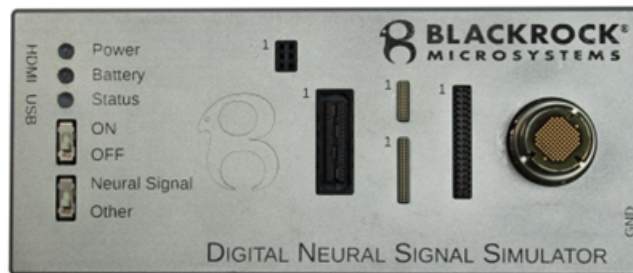
The following analog headstages are currently available:

- 64-Channel Splitter Box (1-64)
- 64-Channel Splitter Box (65-128)
- 16-Ch Macro Cabrio Butterfly Adapter
- 32-Ch Cabrio Butterfly Adapter
- Cabrio Butterfly Adapter
- Quad IDC-10 Adapter

8.3 Digital Neural Signal Simulator (DNSS)

The Digital Neural Signal Simulator supplies simulated field potentials and action potentials (spikes) as well as sine waves at different frequencies. It can be used to test a recording system without being connected to a patient or subject.

Figure 8-1: Digital Neural Signal Simulator



9 SOFTWARE

Open Central by selecting the **Central** shortcut on the desktop. The Central main application connects to a Blackrock Neural Signal Processor to configure hardware channels and acquire data. Incoming data can be processed and visualized in various ways before being saved for future analysis.

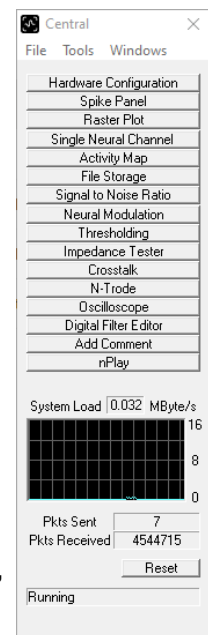
9.1 Central main screen

Select the following options on the **Central** main screen to use the Cerebus Central Suite's functions, features, and settings:

Figure 9-1: Central main screen

- **Hardware Configuration** - Select to open the individual channel configurations and system settings:
 - Go to **Signal Types** or **NSP > Channels** on the tree menu to use the channel sorting options. See **NSP channels** on page 48.
 - Go to **NSP > Settings** on the tree menu to use the following settings:
 - **Global Settings**
 - **LNC Settings**
 - **Filters**
 - **Auto Thresholding**
 - **Spike Sorting**
 - **Adaptive Filtering**
 - **N-Trode Groups**
 - Select **Icon View**, **List View**, **Detail View**, or **Full Row Details** to change the display mode.
 - Right-click a channel and select **Properties**, or double-click a channel, to open its settings. See **Hardware configuration** on page 36.
- **Spike Panel** - Select this option to open the window that graphs and tabulates detected spikes on the neural channels. This screen efficiently visualizes the overall functioning of the system. If spike-sorting rules are defined, the spikes belonging to different units will be displayed in different colors. Double-click any channel to open the **Single Neural Channel Configuration** screen. See **Spike panel** on page 65.
- **Raster Plot** - Select this option to open the screen that graphs the occurrence of spike events on multiple channels over time. This window can show all channels. Continuous traces and comments can also be shown on the plot. To add a comment to the neural events (.nev) file, start typing and complete the **Comment** dialog box that opens. Comments are displayed on the Comment line at the bottom of the **Raster Plot** screen.

*Note: Depending on your current raster plot configuration, you might need to add the comment line to the raster plot view through the channel chooser feature. See **Raster plot** on page 68.*



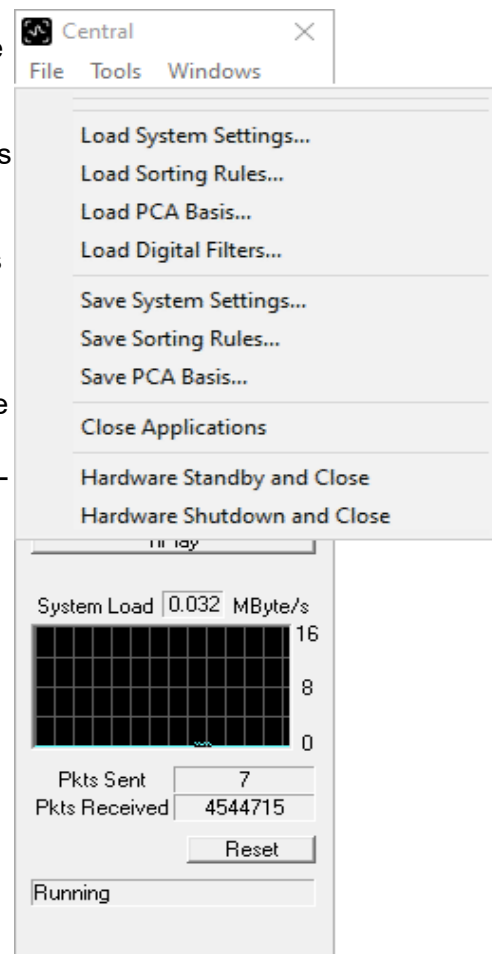
-
- **Single Neural Channel** - Select this option to open the screen that displays the selected channel in a continuous data trace, strip chart, and waves panes. Use this window to define spike-sorting rules and view single-channel activity. See **Single Neural Channel screen** on page 70.
 - **Activity Map** - Select this option to open the color-coding options for the firing rate of neurons recorded by each electrode on a two-dimensional grid. An electrode map file dictates the spatial arrangement of channels. Use the **Activity Map** screen to visualize spike rates over all channels. See **Activity map** on page 82.
 - **File Storage** - Select this option and use the screen that opens to start or stop recording, and to edit additional settings, including file type, file name, and file path for recordings. See **File storage** on page 77.
 - **Signal to Noise Ratio** - Select this option to open a screen showing the signal-to-noise ratio for each channel. Use this window to view and record valuable information about signal quality and electrode functionality. See **Signal-to-noise ratio** on page 85.
 - **Neural Modulation** - Select this option to open the screen that shows a histogram of changes in firing frequency over time. The display shows which channels are recording from neurons with variable firing rates. See **Neural modulation** on page 86.
 - **Thresholding** - Select this option to open the global spike detection threshold settings. Enter the threshold as a value of or a multiplier of the root mean squared (RMS) signal energy.
 - **Impedance Tester** - Select this option to open the screen that measures and display the impedance of each electrode. Use this window to assess electrode functionality. See **Impedance tester** on page 86.
 - **Crosstalk** - Select this option to open the screen that measures and displays the amount of signal shared between channels. Use this screen to mark and disable redundant channels. See **Crosstalk** on page 89.
 - **N-Trode** - Select this option to open the screen that configures and visualizes N-Trode groups. The **Spike Display** in the upper pane shows spikes on each channel. The **Peak/Valley** dialog box is used to define spike units for the group, and the **Continuous Data Display** shows the full signal from the group. See **N-Trode** on page 89.
 - **Oscilloscope** - Select this option to open a screen that graphs the signal acquired by any channel. Use this window to view, measure, and compare signals with custom triggers. See **Oscilloscope** on page 91.
 - **Digital Filter Editor** - Select this option to open a MATLAB utility and configure the custom digital filters. See **Digital filter editor** on page 94.
 - **Add Comment** - Select this option and use the **Comment** dialog box to add a timestamped text comment in the neural events (.nev) file.
 - **nPlay** - Select this option and use the screen that opens to load and replay saved data files for analysis and rerecording.
 - **System Load** - This area graphs the volume of incoming data (MBps) from the Neural Signal Processor.

9.1.1 File menu

The **File** menu provides the following options:

Figure 9-2: File menu on the Central main screen

- **Load System Settings** - Select this option to load the channel settings in CCF and RCF file formats.
- **Load Sorting Rules** - Select this option to load system-calculated Histogram Peak Count sorting rules in CSR format.
- **Load PCA Basis** - Select this option to load system-calculated principal component analysis basis vectors in CFSD format.
- **Load Digital Filters** - Select this option to load digital filters created by the Digital Filter Editor (XML format).
- **Save System Settings** - Select this option to save the current Hardware Configuration channel settings.
- **Save Sorting Rules** - Select this option to save spike-sorting parameters.
- **Save PCA Basis** - Select this option to save PCA basis vectors that are calculated during spike sorting.
- **Close Applications** - Select this option to close Central programs while leaving the hardware running.
- **Hardware Standby and Close** - Select this option to close Central programs and place the hardware in standby mode. The system must be in standby mode for firmware updates.
- **Hardware Shutdown and Close** - Select this option to close Central and its associated applications and shut down the NSP. Turn off the system power switch manually to shut down the hardware.

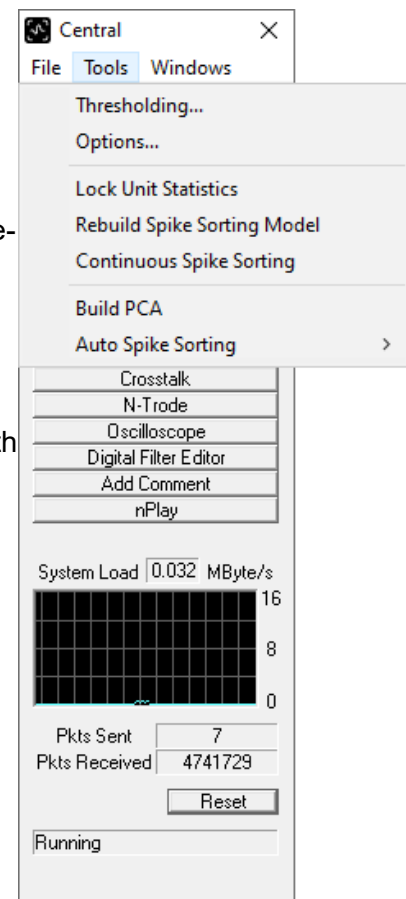


9.1.2 Tools menu

The main **Tools** menu provides several spike sorting options.

Figure 9-3: Tools menu

- **Thresholding** - Set the manual threshold level for all channels as an RMS multiplier or a fixed voltage value.
- **Options** - See below.
- **Lock Unit Statistics** - Disables continuous updates of spike sorting rules.
- **Rebuild Spike-sorting Model** - Restarts the automatic spike-sorting function.
- **Continuous Spike Sorting** - Selecting this item will continuously adjust the spike-sorting statistics, including the splitting and merging of units. Avoid using this setting while recording.
- **Build PCA** - Builds the PCA basis vectors for all channels with PCA based spike sorting.
- **Assisted Spike Sorting** - Enable k-means, DBSCAN, or all PCA methods for spike sorting.

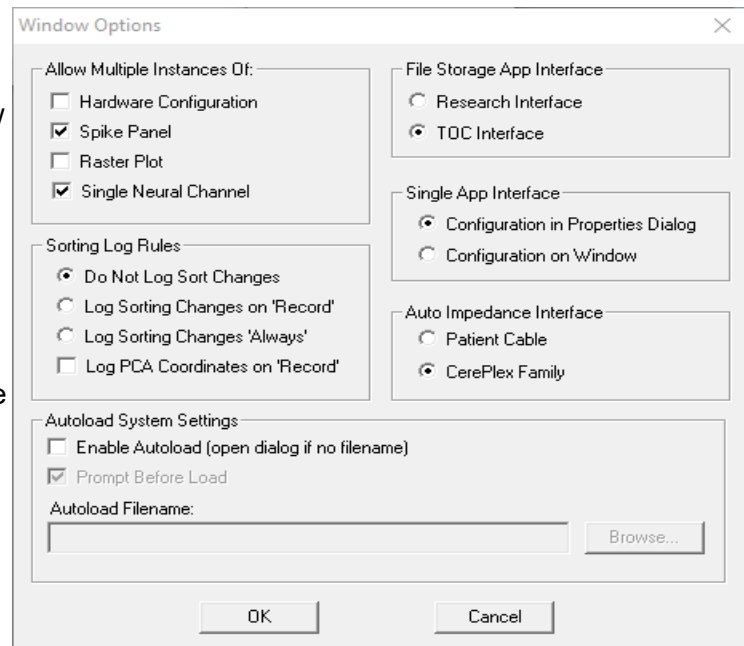


9.1.3 Options

Use the **Window Options** dialog box to configure the general settings for Central.

Figure 9-4: Central Options

- **Allow multiple instances of -** Select this option to allow multiple instances of each selected window type to be opened.
- **Sorting Log Rules -** Sorting rules are manually or automatically defined spike unit classifications that are used to recognize and separate neural spike events.
 - **Do Not Log Sort Changes** - Select this option to disable sort change logging.
 - **Log Sorting Changes on 'Record'** - Select this option to automatically log the spike-sorting model, the signal to noise ratio summary, and the crosstalk summary if it is rebuilt during recording.
 - **Log Sorting Changes 'Always'** - Select this option to log the spike-sorting model, the signal to noise ratio summary, and the crosstalk summary whenever it is rebuilt.
 - **Log PCA coordinates on 'Record'** - Select this option to log the spike-sorting model only when recording begins.
- **File Storage Interface** - Select an option to save files in the Research interface or the TOC interface.
 - **Research interface** - Select for research settings where files are manually named. See **Research interface** on page 77.
 - **TOC interface** - Select for clinical setting where patient information is entered.
- **Single App Interface** - Select a channel configuration method.
 - **Configure in Properties Dialog** - Select to change the channel configuration in Single Neural Channel viewer.
 - **Configure on Window** - Select to change the channel configuration in the **Edit > Properties** menu.
- **Auto Impedance Interface** - Select this option to configure the impedance testing for digital headstages.



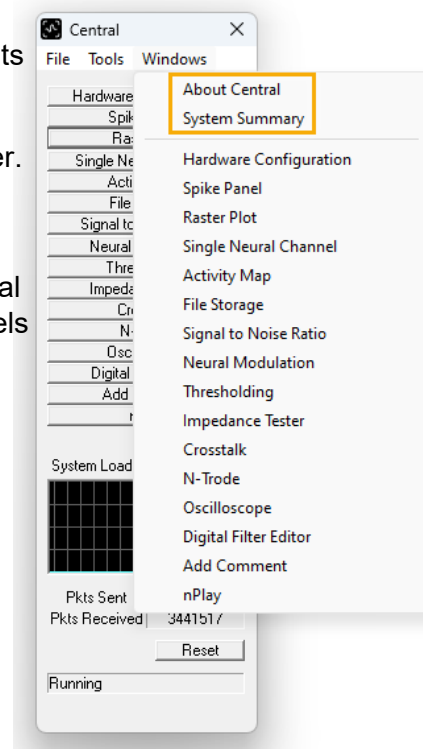
- **CerePlex Family** - Select to configure the impedance testing for digital headstages.
- **Autoload System Settings**
 - **Enable Autoload (opens dialog if no filename)** - Select to open the selected file. If this is selected but no file is entered below, a dialog box will open to select a file.
 - **Prompt before load** - Select to have Central prompt the user to select a CCF file at startup.
 - **Autoload Filename** - Enter the CCF file to be loaded.
 - **Browse** - Click to select a CCF file from the Host PC.

9.1.4 Windows menu

The **Windows** menu summarizes system and software summary information and provides options for Central applications.

- **About Central** - Displays the software version and the system firmware version. Provide this information to support.
- **System Summary** - Displays the average number of units per channel when the automatic histogram peak-count sorting is enabled. It also provides options to disable crosstalking channels and to launch the impedance tester.

Figure 9-5: Windows menu

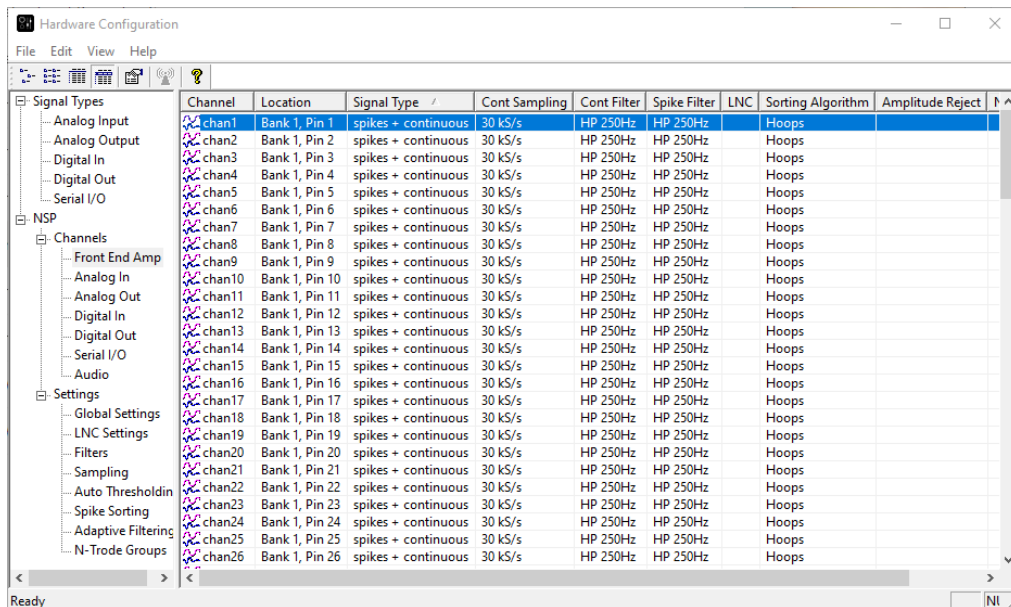


9.2 Hardware configuration

Use the **Hardware Configuration** screen to configure individual channels and system settings. The main panel displays channels in an icon or a list view as selected on the tool bar.

1. Double-click the selected channels to open the **Channel Properties** window for editing.
2. Click the categories in the left window pane to filter the channels.
3. Click **Settings** in the left pane to access additional settings.

Figure 9-6: Hardware Configuration screen



9.2.1 Configuring channel properties

To configure the channel properties:

1. Go to the **Hardware Configuration** screen.
2. Select one or more channels in the right pane.
3. Right-click the selection of channels, and then choose **Properties** or select **Channels** on the toolbar.

Each type of channel and its associated options are described below.

9.2.2 Signal Type configurations

9.2.2.1 Analog Neural Inputs

To configure the hardware settings for one or more analog inputs:

1. On the **Central** main screen, select **Hardware Configuration**.
2. On the **Hardware Configuration** screen that opens, select **Signal Types > Analog Input** in the left pane.
3. Select one or more channels of interest.
4. Right-click your selection and select **Properties**.
5. Use the **Analog Input Properties** screen that opens to configure the hardware settings for the selection of analog inputs. There are three sections on this screen : **Location**, **Continuous Acquisition**, and **Enable Spike Processing** (see below).

Figure 9-7: Analog Input Properties screen with one channel selected

The screenshot shows the 'Analog Input Properties' dialog box with the following settings:

- Location:** Bank: Front End Pin: 2
- Label:** chan2
- Digital Range:** -32764 to 32764
- Analog Range:** -8191 to 8191
- Analog Units:** μV
- External Gain:** 1
- ☐ Enable Line Noise Cancellation
 - Estimate Time Constant (sec): 10
 - LNC Method: Soft Synch
- Continuous Acquisition:**
 - Filter: <none>
 - Graphical Chooser
 - Sampling Rate: <none>
 - Channel Chooser
- ☐ Raw Data
 - ☐ Enable (Will remove 30kS/s if enabled)
 - ☒ AC Input Coupling
 - ☐ Enable Software Referencing
 - Ref Elec:
 - Type:
- ☒ Enable Spike Processing
 - Filter: HP 250Hz
 - Graphical Chooser
 - Threshold Value (μV): -63
 - RMS Threshold
 - Sorting Algorithm: Hoops
 - ☐ Enable Amplitude Rejection
 - Upper limit (μV): 8191
 - Lower limit (μV): -8191

Buttons at the bottom: Raster View, Spike View, OK, Cancel, Apply.

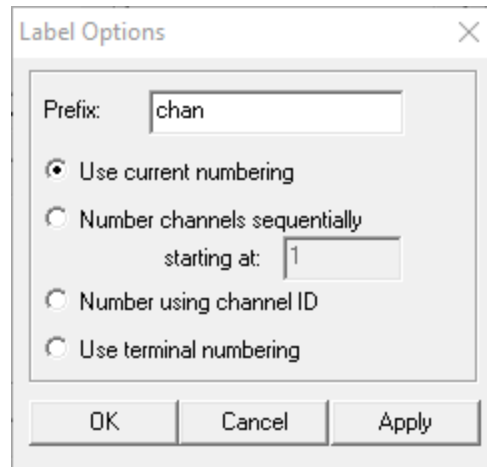
Figure 9-8: Analog Input Properties screen with multiple channels selected

The screenshot shows the 'Analog Input Properties' dialog box with the following settings:

- Location:** Bank: Front End
- Label:** Edit
- Digital Range:** -32764 to 32764
- Analog Range:** -8191 to 8191
- Analog Units:** μV
- External Gain:** 1
- ☐ Enable Line Noise Cancellation
 - Estimate Time Constant (sec): 10
 - LNC Method: Soft Synch
- Continuous Acquisition:**
 - Filter: HP 250Hz
 - Graphical Chooser
 - Sampling Rate: 30 kS/s
 - Channel Chooser
- ☐ Raw Data
 - ☐ Enable (Will remove 30kS/s if enabled)
 - ☒ AC Input Coupling
 - ☐ Enable Software Referencing
 - Ref Elec:
 - Type:
- ☒ Enable Spike Processing
 - Filter: HP 250Hz
 - Graphical Chooser
 - Threshold Value (μV): -63
 - RMS Threshold
 - Sorting Algorithm: Hoops
 - ☐ Enable Amplitude Rejection
 - Upper limit (μV): 8191
 - Lower limit (μV): -8191

Buttons at the bottom: Raster View, Spike View, OK, Cancel, Apply.

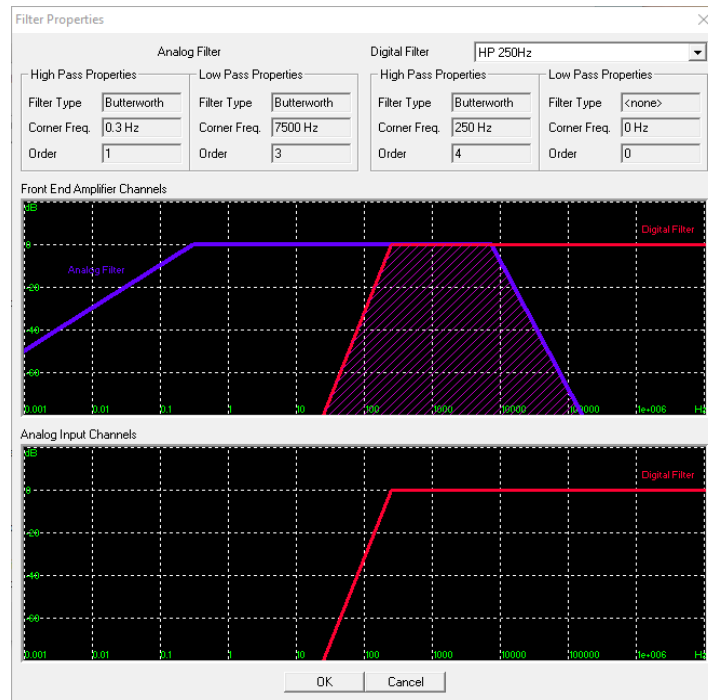
- **Location**
 - **Label** - If you selected only one channel, enter a user-defined name. If you selected multiple channels, select **Edit** and use the **Edit Multiple Labels** dialog box to automatically assign labels to them.

Figure 9-9: Edit Multiple Labels dialog box

- **Digital Range** - Fixed entry showing the range of digital values that analog signals are mapped to.
- **Analog Range** - Fixed entry showing the range of analog values measured.
- **Analog Units** - Unit of analog signal measurement displayed for reference.
- **External Gain** - Gain value of the headstage. All Blackrock systems use a gain of 1.
- **Enable Line Noise Cancellation** - Select to enable cancellation of 50 or 60 Hz noise.
 - **Estimate Time Constant** - The period over which line noise parameters are determined.
 - **LNC Method** - Select software or hardware line noise cancellation.
- **Continuous Acquisition**

- **Filter**- Select a digital filter to apply to continuous data.

Figure 9-10: Filter Properties dialog box



- **Graphical Chooser** - Select this option to visualize the continuous signal filter that is currently applied.
- **Sampling Rate** - Select the sampling rate of data acquisition.
- **Channel Chooser** - Select the sampling rate for data acquisition on any channel.
- **Raw Data** - Record raw data stream before filtering or sorting is applied.
- **AC Input Coupling** - High pass filter to remove DC offset.
- **Enable Software Referencing** - Apply software referencing to local field potential or both Local field potential and spike data.
 - **Reference Electrode** - Select channel to use as software reference.
 - **Type** - Select whether software referencing should be applied to local field potentials only or local field potentials and spike data.
- **Enable Spike Processing** - Select this option to allow the detection of spike events on channel.
 - **Filter** - Filter selection for spike event processing.
 - **Graphical Chooser** - Select this option to visualize the spike processing filter that is currently applied. See **Figure 9-10: Filter Properties dialog box**.
 - **Auto** - Select this option to apply automatic thresholding that does not rely on the hard coded voltage value computed from either an RMS voltage multiplier or user-defined voltage. Auto-thresholding algorithmically adjusts the threshold voltage value based on

the signal-to-noise ratio of the incoming signal. See **Auto thresholding** on page 53, in particular **Figure 9-29: Auto Thresholding Properties dialog box**.

Note: When this option is selected an algorithm detects spikes outside of the noise band. You can adjust the thresholding on the Single Neural Channel or the Thresholding applications

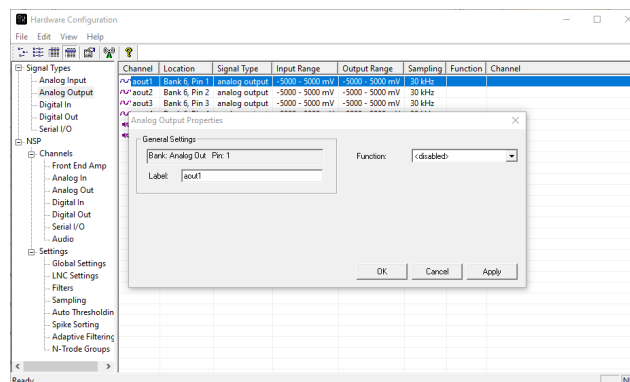
- **Threshold Value** - Voltage threshold for classification of a spike event in mV.
- **RMS Threshold** - Set parameters for auto-threshold calculation.
- **Sorting Algorithm** - Set the sorting algorithm for a channel as described in "**Spike sorting**" on page 54.
- **Enable Amplitude Rejection** - Rejects waveforms exceeding a specified amplitude.
 - **Upper Lower Limit (uV)** - Voltage upper limits for amplitude rejection.
 - **Lower Lower Limit (uV)** - Voltage lower limits for amplitude rejection.
- **Raster View** - Selecting this option opens the **Raster Plot** screen with the selected channel highlighted.
- **Spike View** - Selecting this option opens the **Single Neural Channel Configuration** screen for the selected channel.

9.2.2.2 Analog output

To configure the analog output ports:

1. On the **Hardware Configuration** screen, select **Signal Types> Analog Output**.
2. Right-click a channel of interest and select **Properties**.
3. Use the **Analog Output Properties** screen that opens to configure the analog output ports.

Figure 9-11: Analog Output Properties screen



Analog outputs can function as monitor channels, directly mirroring signals of interest, or as waveform generators. Each modality has the following output and triggering parameters:

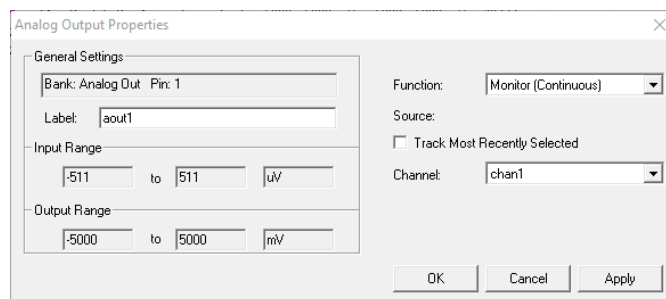
- **Label** - Shows the user-defined channel label.
- **Function** - Select the function to be assigned to an analog channel.

Analog output: continuous monitor

An analog output channel configured as a continuous monitor will scale a user input signal into a plus or minus 5 V output signal.

Figure 9-12: Analog Output Properties for Continuous Monitor

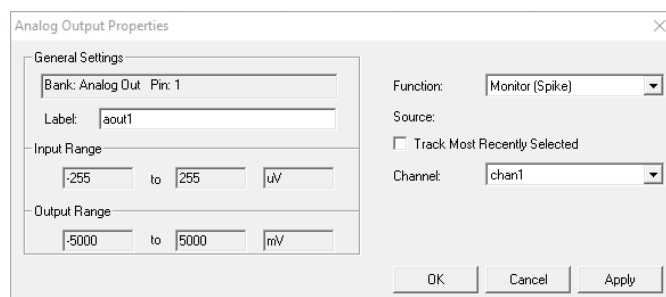
- **Function** - Select the **Monitor (Continuous)** analog output option.
- **Input Range** - Displays the input voltage range.
- **Output Range** - Displays the output voltage range.
- **Source: Track Most Recently Selected** - Select this check box to use the channel that is selected on the **Spike Panel**, **Single Neural Channel**, or **Raster Plot** screens as the monitor input.
- **Channel** - Select a channel. An analog output channel configured as a spike monitor will scale a user selected input signal into a plus or minus 5 V output signal.

*Analog output: spike monitor*

An analog output channel configured as a spike monitor will scale a user selected input signal into a plus or minus 5 V output signal containing only identified spike events.

Figure 9-13: Analog Output Properties for Spike Monitor

- **Function** - Select the **Monitor (Spike)** analog output option.
- **Input Range** - Displays the input voltage range.
- **Output Range** - Displays the output voltage range.
- **Source: Track Most Recently Selected** - Select this check box to use the channel that is selected on the **Spike Panel**, **Single Neural Channel**, or **Raster Plot** screens as the monitor input.
- **Channel** - Select a channel. An analog output channel configured as a spike monitor will scale a user selected input signal into a plus or minus 5 V output signal.

*Analog output: sine waveform*

An analog output channel configured as a sine waveform generator will produce a user-defined sine wave on a user-defined trigger stimulus.

Figure 9-14: Analog Output Properties for Sine Waveform

- **Function** - Select the **Sine Waveform** analog output option.
- **Amplitude (mV)** - Enter the amplitude of the output sine wave.
- **Frequency (Hz)** - Enter the frequency of the output sine wave.
- **Offset (mV)** - Enter the DC offset of the output sine wave.
- **Repeats** - Enter the number of times to repeat the waveform after it is triggered. A value of 0 will repeat the waveform indefinitely.
- **Wave Number** - Select one of five waveforms.
- **Trigger** - Select a triggering option. For more information, see **Analog output: triggering** on page 44.

The screenshot shows the 'Analog Output Properties' dialog box. It has a title bar with a close button. The dialog is divided into two main sections. The left section, titled 'General Settings', contains a 'Bank' field with the value 'Analog Out Pin: 1' and a 'Label' field with the value 'aout1'. The right section contains a 'Function' dropdown menu set to 'Sine Waveform', a 'Wave number' dropdown menu set to '1', and a 'Trigger' dropdown menu set to 'Instant'. Below these are four input fields: 'Amplitude(mV):' with the value '0', 'Frequency(Hz):' with the value '60', 'Offset (mV):' with the value '0', and 'Repeats:' with the value '0'. At the bottom of the dialog are five buttons: 'Load', 'Save', 'OK', 'Cancel', and 'Apply'.

Analog output: custom waveform

An analog output channel configured as a custom waveform generator will produce a user-defined waveform on a user-defined trigger stimulus.

Figure 9-15: Analog Output Properties for Custom Waveform

- **Function** - Select the **Custom Waveform** analog output option.
- **Wave Number** - Select one of the five waveforms.
- **Trigger** - Select a triggering option. See **Analog output: triggering** below.
- **Phase Settings** - Select one of the following options:
 - **Number of Phases** - Select the number of voltage phases in the waveform (1-248).
 - **Offset (mV)** - Enter the DC offset of the waveform (millivolts).
 - **Phase** - Select the phase that will be used.
 - **Repeats** - Enter the number of times to repeat the waveform after it is triggered. A value of 0 will repeat the waveform indefinitely.
 - **Amplitude (mV)** - Enter the phase amplitude (millivolts).
 - **Duration (ms)** - Enter the phase duration (milliseconds).

Analog Output Properties

General Settings

Bank: Analog Out Pin: 1

Label: aout1

Function: Custom Waveform

Wave number: 1

Trigger: Instant

Cycle Length (ms): 2.0

Frequency (Hz): 500.00

Output Range (mV): -5000 to 5000

Phase Settings

Number of Phases: 2

Offset (mV): 0

Repeats: 0

Phase: No. 1

Amplitude (mV): 0

Duration (ms): 1

5000

mV

-5000

0

ms

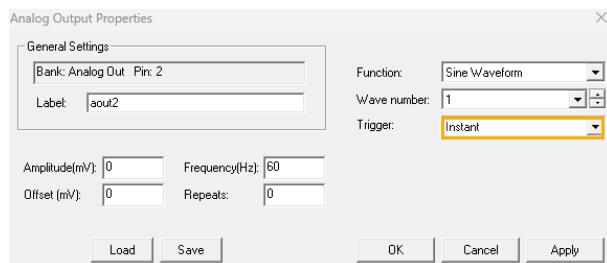
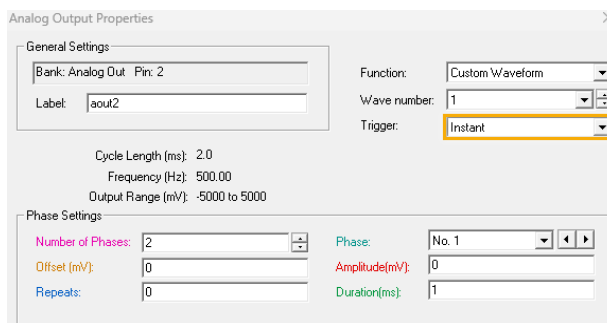
2.0

Reset Load Save OK Cancel Apply

Analog output: triggering

Sine and custom waveforms can be triggered by the following conditions:

- **Instant** - Waveform triggers when **OK** or **Apply** are selected.
- **Digital Input Rising Edge** - Waveform triggers when a user-defined digital input pin transitions from low to high. Select a digital input bit.
- **Digital Input Falling Edge** - Waveform triggers when a user defined digital input pin transitions from high to low. Select a digital input bit.
- **Spike Unit** - Waveform triggers when a sorted unit is detected on a defined channel.
- **NeuroMotive ROI** - Triggers when a subject enters or exits a selected region of interest.
- **Recording Start** - Triggers when a recording is started.

Figure 9-16: Triggering options for sine waveforms**Figure 9-17: Triggering options for custom waveforms**

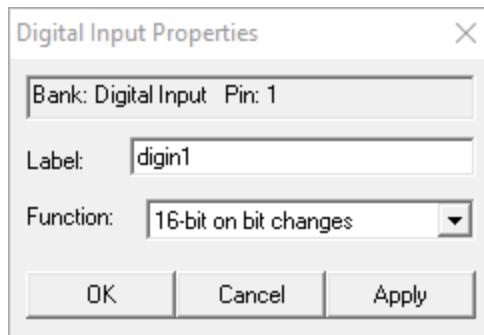
Note: If you select **Extension** in **Function**, the waveform triggers based on custom extension codes loaded on to the hardware instead of an option you select in **Trigger**.

9.2.2.3 Digital input

To edit the digital input settings:

1. On the **Hardware Configuration** screen, select **Signal Types > Digital In** to view a list of input channels .
2. In the list, right-click a channel of interest and select **Properties**.
3. Use the **Digital Input Properties** dialog box to configure the digital input port.

Note: Digital inputs can synchronize external hardware and pass digital data into neural recordings.

Figure 9-18: Digital Input Properties dialog box

Select an option in **Function** :

- **16-bit on word strobe** - Reads 16 bits on the rising edge of the strobe pin.
- **16-bit on bit changes** - 16-bit read when any bit changes from LOW to HIGH or from HIGH to LOW.
- **16-bit on rising edge** - 16-bit read when any bit changes from LOW to HIGH. Each bit indicates if a LOW to HIGH change occurred on each digital pin.
- **16-bit on falling edge** - 16-bit read when any bit changes from HIGH to LOW. Each bit indicates if a HIGH to LOW change occurred on each digital pin.
- **8-bit strobe / 8-bit bit changes** - Reads bits 0-7 on the rising edge of the strobe pin and reads bits 8-15 if bits 8-15 changed from LOW to HIGH or HIGH to LOW.
- **8-bit strobe / 8-bit bit rising edge** - Reads bits 0-7 on the rising edge of the strobe pin, and bits 8-15 indicate if digital pins D8-D15 changed from LOW to HIGH.
- **8-bit strobe / 8-bit bit falling edge** - Reads bits 0-7 on the rising edge of the strobe pin, and bits 8-15 indicate if digital pins D8-D15 changed from HIGH to LOW.

9.2.2.4 Digital Output

Click **Signal Types** and then click **Digital Out** on the **Hardware Configuration** screen to open the **Digital Output Properties** screen and configure the properties of the digital outputs. Digital outputs can function as spike detection monitors, timed function generators, or triggered events.

Digital output: Monitor Electrode

A digital output in monitor electrode mode will send a high pulse when a sorted spike unit is detected on a defined channel. Use the dialog box shown below to configure the digital output settings.

Figure 9-19: Digital Output Properties: Monitor Electrode

- **Properties** - Select **Monitor Electrode**.
- **Track Most Recently Selected** - Select this check box to choose a channel on the Spike Panel, Single Neural Channel, or Raster Plot screen to use as the monitor input.
- **Source** - Select the channel to monitor.
- **Monitored Units** - Select each spike unit classification that will be a trigger.

The screenshot shows the 'Digital Output Properties' dialog box. Under 'General Settings', 'Bank' is 'Digital Out' and 'Pin' is '1'. 'Label' is 'dout1'. Under 'Properties', 'Monitor Electrode' is selected in the dropdown. The 'Track Most Recently Selected' checkbox is unchecked. 'Source' is '<none>'. Under 'Monitored Units', a table shows 'UnClassified' and units 1 through 5, all with checked checkboxes.

UnClassified	1	2	3	4	5
☒	☒	☒	☒	☒	☒

Digital output: Timed

A digital output in timed mode will send a pulse train with specified parameters.

Use the dialog box shown below to configure the signal.

Figure 9-20: Digital Output Properties: Timed

- **Properties** - Select **Timed**.
- **Frequency (Hz)** - Move the slider to change the number of pulses per second.
- **Duty Cycle** - Move the slider to the percentage of time a pulse train is high.
- **Samples High** - Enter the number of 30kHz sample periods output should be held high.
- **Samples Low** - Enter the number of 30kHz sample periods output should be held low.
- **Offset (mV)** - Enter the DC offset voltage of the digital output.

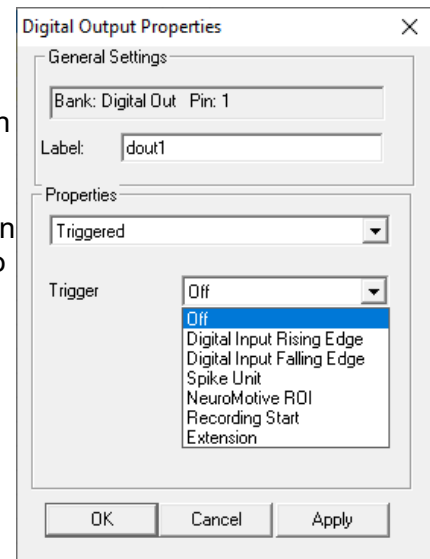
The screenshot shows the 'Digital Output Properties' dialog box. Under 'General Settings', 'Bank' is 'Digital Out' and 'Pin' is '1'. 'Label' is 'dout1'. Under 'Properties', 'Timed' is selected in the dropdown. 'Frequency' is set to 15000Hz with a slider. 'Duty Cycle' is set to 50% with a slider. 'Samples High' is 1, 'Samples Low' is 1, and 'Samples Offset' is 0.

Digital output: Triggered

A digital output in triggered mode will send a pulse when the selected trigger stimulus occurs.

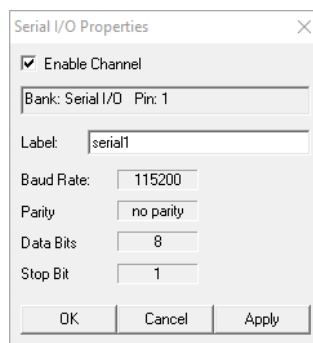
Figure 9-21: Digital Output Properties: Triggered

- **Properties** - Select **Triggered**.
- **Trigger** - Select a trigger type:
 - **Digital Input Rising Edge** - Waveform triggers when a user defined digital input pin transitions from low to high. A digital input bit must be selected.
 - **Digital Input Falling Edge** - Waveform triggers when a user defined digital input pin transitions from high to low. A digital input bit must be selected.
 - **Spike Unit** - Waveform triggers when a sorted unit is detected on a defined channel.
 - **NeuroMotive ROI** - Waveform triggers when a subject enters or exits a region of interest.
 - **Recording Start** - Waveform triggers when a recording begins.
 - **Extension** - Waveform triggers based on custom extension code loaded onto hardware.



9.2.2.5 Serial I/O

The **Serial I/O Properties** dialog box displays the serial communication parameters and allows you to name and enable the serial port. Serial communication parameters cannot be changed.

Figure 9-22: Serial I/O dialog box

9.2.3 NSP channels

Central supports the following channels (see the figure below):

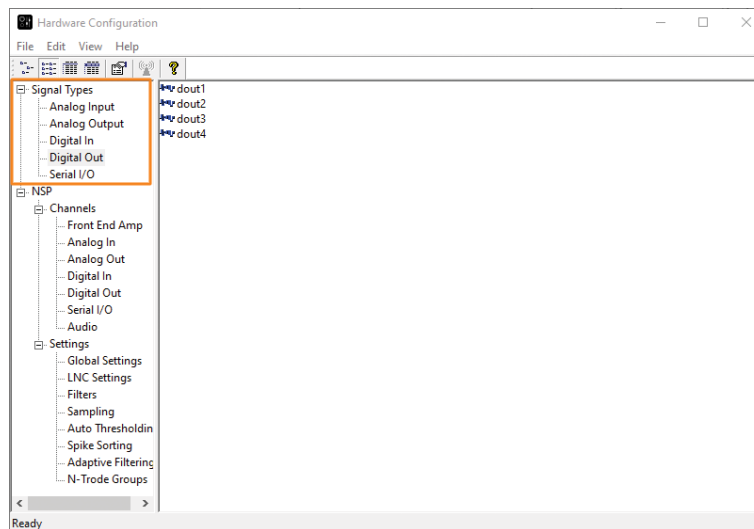
- Neural recording inputs (labeled Front-End Amplifier inputs in the software)
- Analog inputs and outputs (including audio)
- Digital inputs and outputs
- Serial inputs and outputs

To change the display mode, select an option on the toolbar:

- **Icon View** - Channels are displayed as icons.
- **List View** - Channels are displayed in a tabulated list.

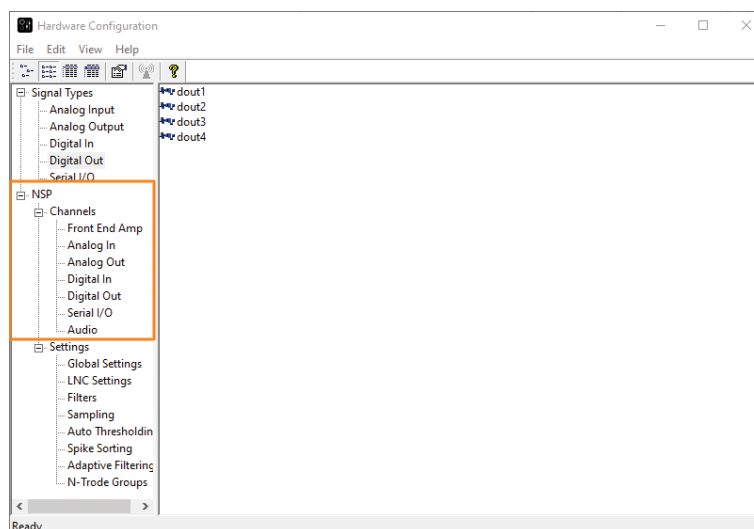
To filter channels by function, go to **Signal Types** on the tree menu, and then select a filtering option. For example, select **Signal Types > Digital Out** to narrow the list of channels by the digital output location.

Figure 9-23: Hardware Configuration screen showing signal type sorting options



To filter channels by a physical location, go to **NSP > Channels**, and then select a filtering option. For example, select **NSP > Channels > Digital Out** to narrow the list of channels by the digital output function.

Figure 9-24: Hardware Configuration screen showing NSP channel sorting options



9.2.4 NSP settings

9.2.4.1 Global settings

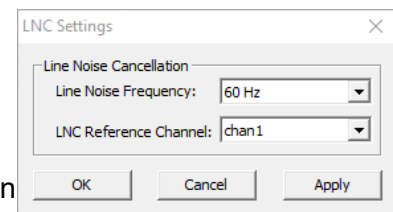
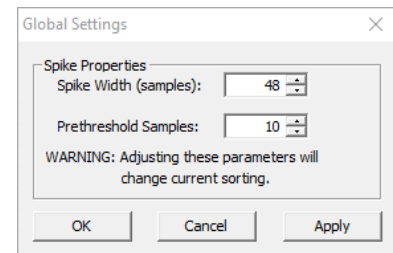
Figure 9-25: Global Settings dialog box

- **Spike Width (samples)** - The number of samples recorded around a detected spike event.
- **Prethreshold Samples** - The number of samples to record preceding a threshold crossing.

9.2.4.2 LNC Settings

Figure 9-26: LNC Settings dialog box

- **Line Noise Frequency (Hz)** - The line frequency of the system, either 50 Hz or 60 Hz.
- **LNC Reference Channel** - The reference channel used to compute system line noise. The extracted line noise template for each channel can be viewed in Single Neural Channel when the Raw/LNC button is checked.

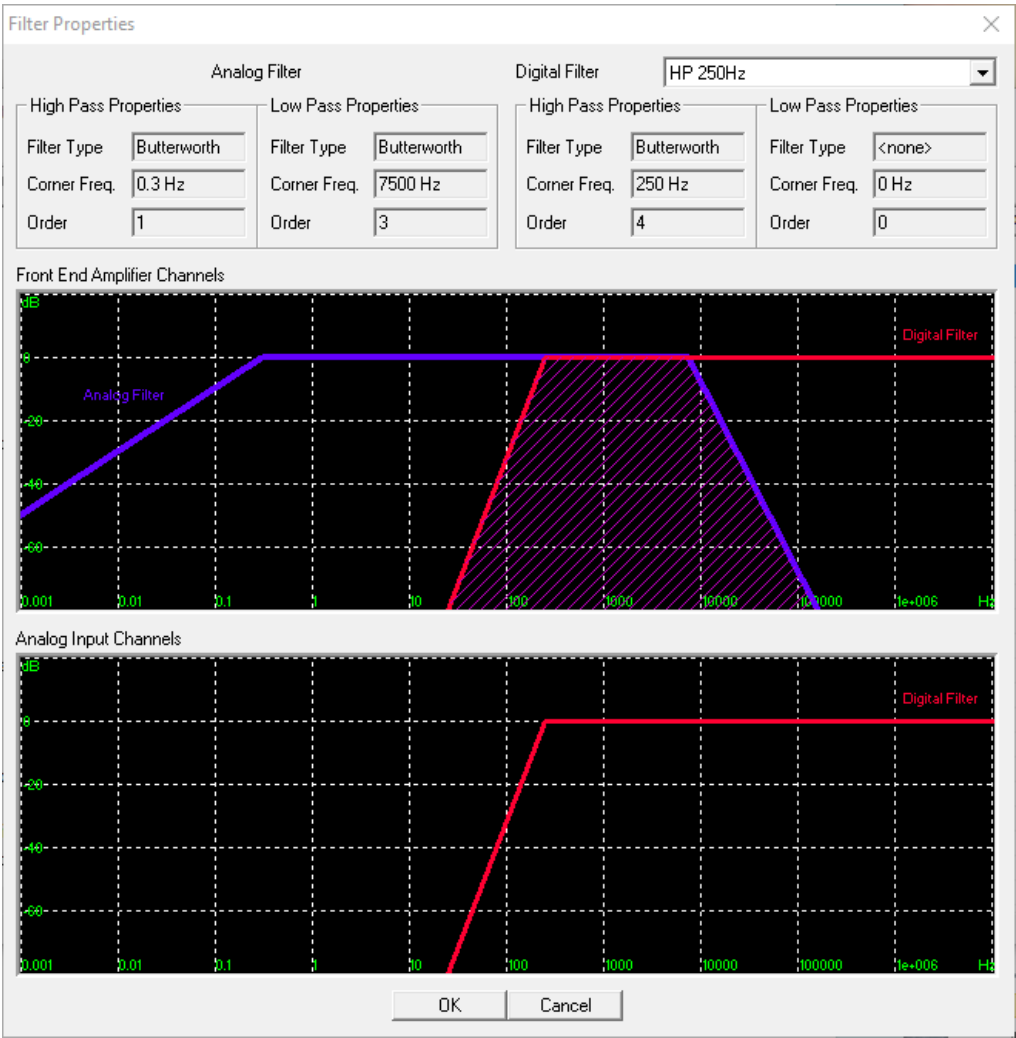


9.2.4.3 Filters

Central provides preset high-pass, low-pass, and band-pass filters.

The **Filter Properties** screen displays the details of a selected digital filter in a bode plot and key numerical parameters.

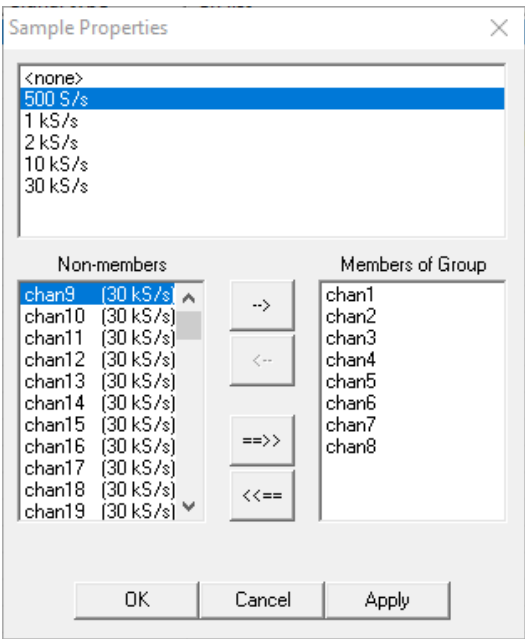
Figure 9-27: Filter Properties screen



9.2.4.4 Sampling

Sampling provides a combined interface dedicated to visualizing and assigning continuous data sampling rates.

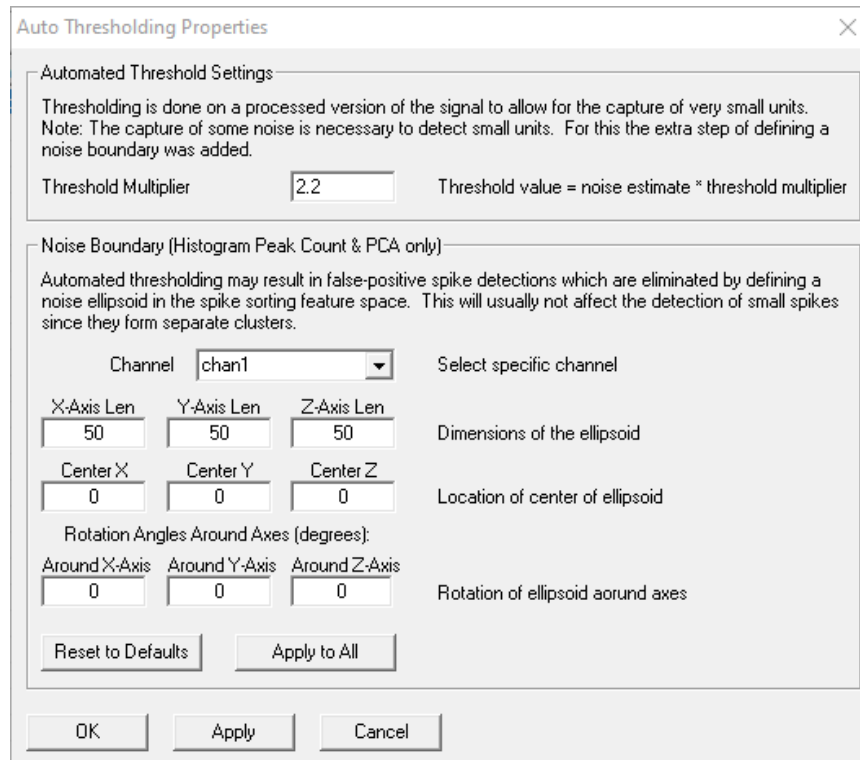
Figure 9-28: Sampling Properties screen



9.2.4.5 Auto thresholding

Use the dialog box shown below to specify the auto thresholding properties for spike detection:

Figure 9-29: Auto Thresholding Properties dialog box



The dialog box is titled "Auto Thresholding Properties" and contains the following sections:

- Automated Threshold Settings:**

Thresholding is done on a processed version of the signal to allow for the capture of very small units. Note: The capture of some noise is necessary to detect small units. For this the extra step of defining a noise boundary was added.

Threshold Multiplier: Threshold value = noise estimate * threshold multiplier
- Noise Boundary (Histogram Peak Count & PCA only):**

Automated thresholding may result in false-positive spike detections which are eliminated by defining a noise ellipsoid in the spike sorting feature space. This will usually not affect the detection of small spikes since they form separate clusters.

Channel: Select specific channel

X-Axis Len: Y-Axis Len: Z-Axis Len: Dimensions of the ellipsoid

Center X: Center Y: Center Z: Location of center of ellipsoid

Rotation Angles Around Axes (degrees):

Around X-Axis: Around Y-Axis: Around Z-Axis: Rotation of ellipsoid around axes

Buttons:

At the bottom are three buttons:

- **Threshold Multiplier** - To register as a spike event, signal energy between 1 and 5 kHz must exceed the energy of the noise by a factor of the threshold multiplier level you enter in this box.
- **Channel** - Select a channel of interest and then apply the following noise boundary properties to this channel:
 - **Dimensions of the Ellipsoid** - Enter the radius of the noise ellipsoid along each axis.
 - **Center of Ellipsoid** - Enter the center of the noise ellipsoid in Cartesian coordinates.
 - **Rotation of Ellipsoid** - Enter the rotation angle of the noise ellipsoid around each axis.
- **Reset to Defaults** - Erases all user-defined values and returns them to factory defaults, as shown in **Figure 9-29: Auto Thresholding Properties dialog box**.
- **Apply to All** - Applies the same settings to all channels, not just the channel selected in the Channel dropdown menu.

9.2.4.6 Spike sorting

The **Spike Sorting Properties** dialog box allows the user to define the rules and configure the settings for the spike sorting and artifact rejection algorithms. The default settings are recommended, but adjusting the settings might improve the performance and accuracy of your specific use case.

Four spike sorting methods are available (see below):

- Manual PCA
- K-means PCA
- DBSCAN PCA
- Histogram Peak Count

The Central software provides automatic and manual methods of real-time spike sorting.

Figure 9-30: PC Space

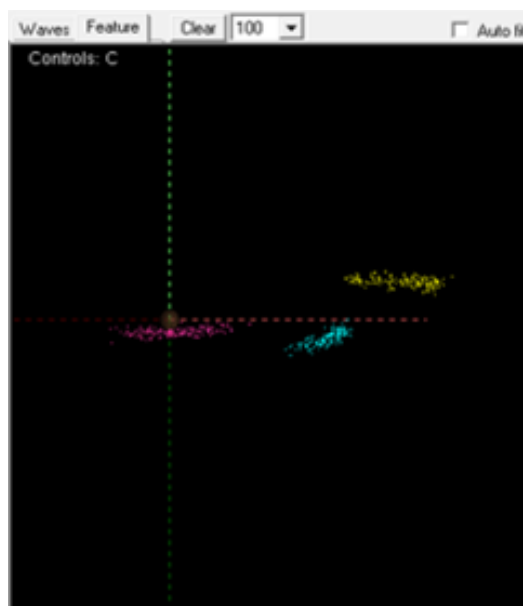
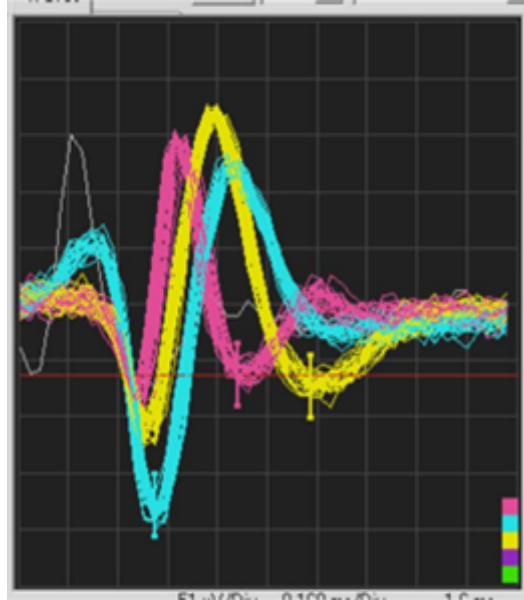


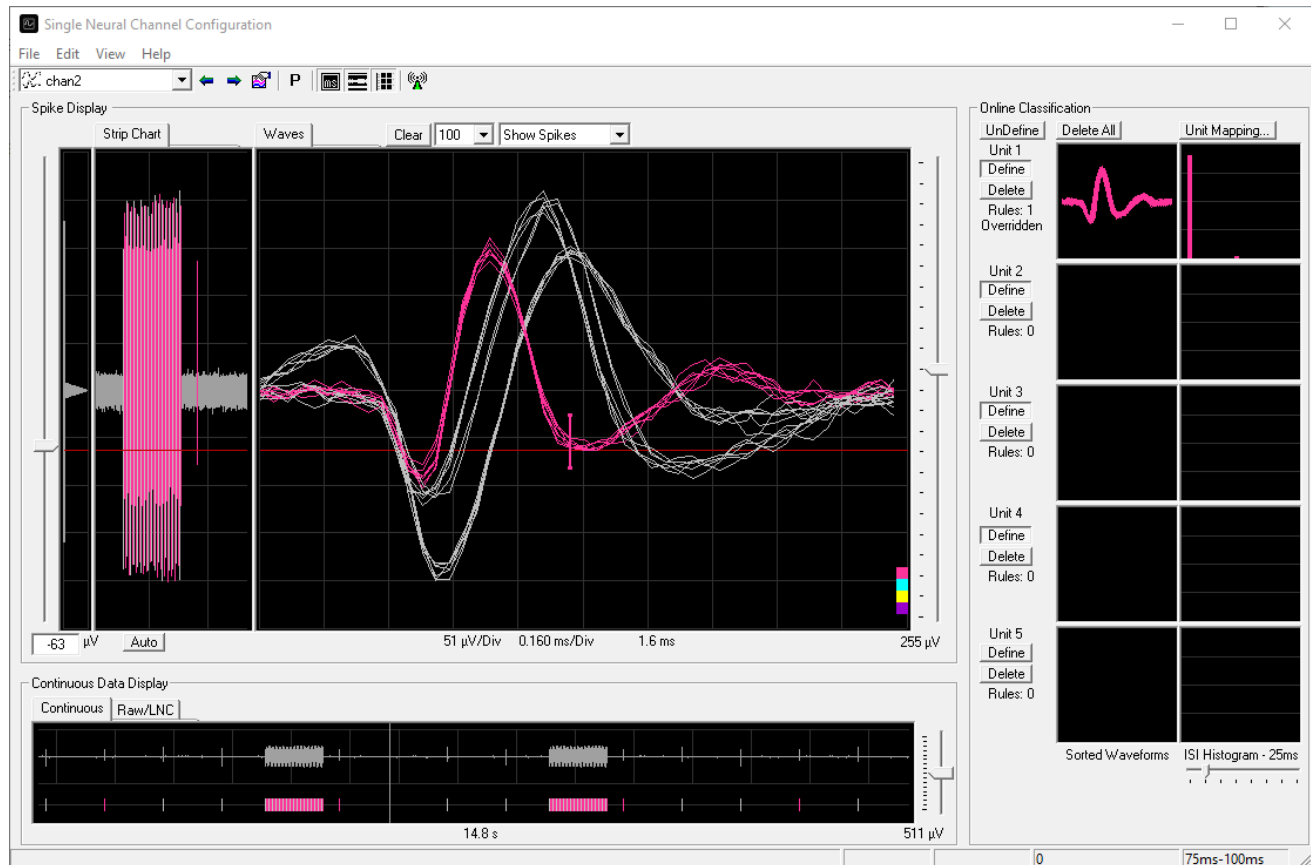
Figure 9-31: Sorted Spikes

Set up histogram peak count (automatic)

1. Open the **Hardware Configuration** window and select channels of interest.
2. Right-click a channel of interest and select **Properties**.
3. Select **Histogram Peak Count** on the spike-sorting menu.
4. Select **OK**.
5. In Central, select **Tools > Rebuild Spike Sorting**. This step may take several minutes and must be repeated for each experimental session. Each time the sorting is rebuilt, new classifications are calculated based on current data.

Note: To save Histogram Peak Counts sorting rules, go to the **Central** main screen and select **File > Save Sorting Rules**.

To load Histogram Peak Counts sorting rules, go to the **Central** main screen and select **File > Load Sorting Rule**.

*Hoops (manual)***Figure 9-32: Single Neural Channel Configuration screen - Hoops**

1. Go to the **Hardware Configuration** screen and select one or more channels of interest.
2. Right-click your selection and select **Properties**.
3. On the **Analog Input Properties** dialog box, select **Hoops** in **Sorting Algorithm**.
4. Go to **Single Neural Channel** and select the channel you want to edit on the top menu bar.

Note: Ensure **View Classification** is selected to display the **Units** pane.

5. Define 1-5 units with 1-4 amplitude bands per channel as follows:
 - On the **Online Classification** pane, select **Define All** to display the color-coded selection handles at the lower right of the **Spike Display** pane.
 - Each selection handle corresponds to a unit:
 - Unit 1 (magenta / pink)
 - Unit 2 (cyan / light blue)
 - Unit 3 (yellow)

■ Unit 4 (purple)

■ Unit 5 (green)

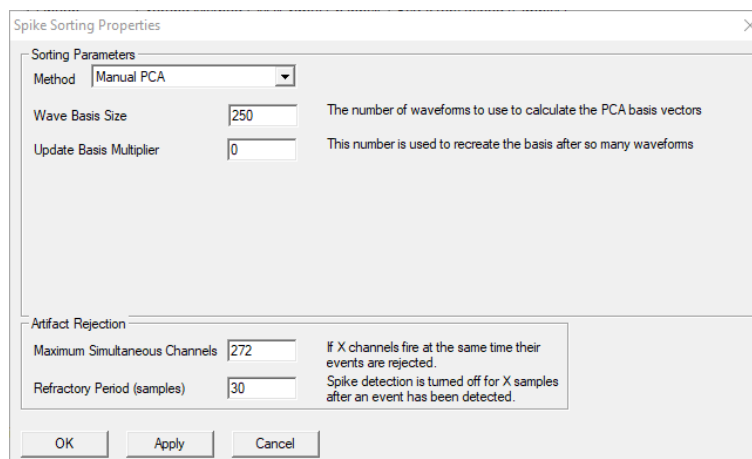
- Drag a selection handle to place an amplitude band on a spike (up to 4 per unit). The **Online Classification** pane displays the spikes and amplitude bands you define. For example, the figure above shows a user defined Unit 1 with 2 amplitude bands (**Rules: 2**).

Note: To save Hoops sorting rules in Central, select **File > Save System Settings**. To load Hoops sorting rules, select **File > Load System Settings**.

Manual PCA

Manual principal component analysis (PCA) uses principal component analysis to generate a feature point for each spike event. To manually define unit classifications, draw an ellipsoid around each cluster of feature points in the feature space.

Figure 9-33: Spike Sorting Properties dialog box - Manual PCA



To set up manual PCA:

1. Go to the **Hardware Configuration** screen and select **NSP > Settings > Spike Sorting**.
2. Right-click **Spike Sorting** and select **Properties**.
3. On the **Spike Sorting Properties** dialog box, change the following settings:
 - **Sorting Parameters**
 - **Method** - Select **Manual PCA**.
 - **Wave Basis Size** - Enter the number of waveforms to use to calculate the PCA basis vectors.
 - **Update Basis Multiplier** - The program recalculates PCA basis vectors after the number of samples exceeds the basis size times the basis multiplier you enter. No recalculations will be made if you enter **0** (zero).
 - **Artifact Rejection**. For more information, see **Artifact Rejection** on page 63
 - Click **OK** to apply the settings and close the dialog box.

- Go to the **Central** main screen and select **Tools > Auto Spike Sorting Method > All PCA Spike Sorting** to sort the channels.

Note: Ensure the channels of interest have been configured to use the Manual PCA sorting algorithm. See **To configure the hardware settings for one or more analog inputs:** on page 37.

- Go to the **Single Neural Channel Configuration** screen and select the channel of interest on the top menu bar. For example, **chan1**.

Note: If the **Online Classification** pane is hidden, select the **View Classification** button on the top menu bar.

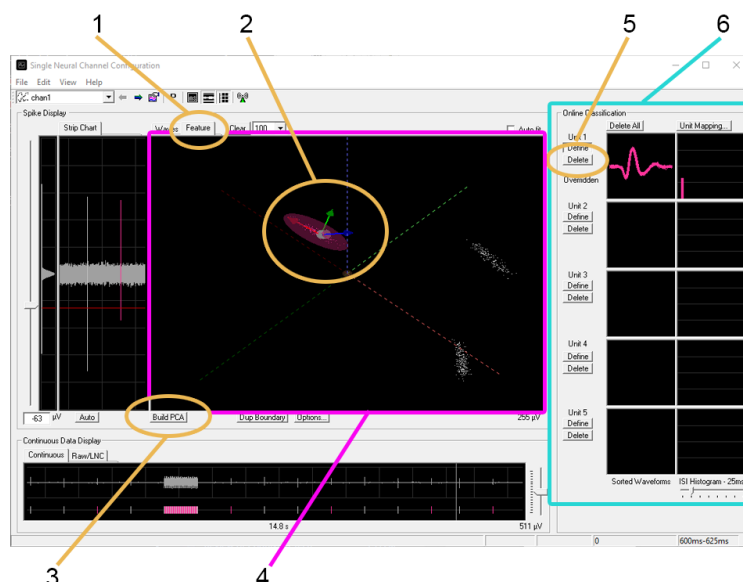
- Select the **Feature** tab and then select **Build PCA**.

Note: A counter at the lower left of the screen shows how many of the total required waveforms have been collected. By default, the program needs 250 waveforms to build the PCA basis vectors. The time required to record 250 waveforms will depend on the spike rates and may take several minutes. To modify the Wave Basis Size (number of waveforms), go to **Hardware Configuration > Spike Sorting**. After the feature points have been generated, clustering accuracy will improve as the number of feature points increases.

- Define each unit as follows:

- On the **Online Classification** pane, select **Define** under **Unit x**. For example, **Unit 1**.
- On the **PCA Space** pane, drag a line around the target data cluster while pressing Ctrl.
- Ensure the data cluster appears inside a shaded ellipsoid on the 3-D graph.

Figure 9-34: Single Neural Channel Configuration screen - Manual PCA



- Feature** button
- A 3-D graph showing a data cluster defined as **Unit 1** inside a shaded ellipsoid
- Build PCA** button

4. **PCA Space** pane
5. **Define** button
6. **PCA Space** pane

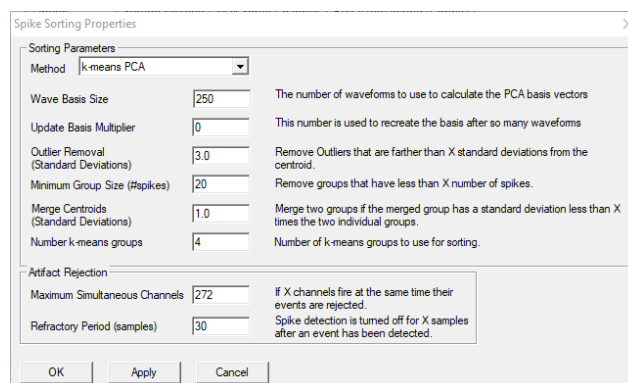
Note:

- To save Manual PCA sorting rules in Central, click on **File > Save System Settings**.
- To load Manual PCA sorting rules, click **File > Load System Settings**.
- To save PCA basis vectors, click **File > Save PCA Basis**.
- To load PCA basis vectors, click **File > Load PCA Basis**.

K-means PCA

K-means principal component analysis (PCA) algorithms automatically generate ellipsoids around k number of clusters to sort recorded units.

Figure 9-35: Spike Sorting Properties - K-means PCA



To set up k-means PCA:

1. Go to the **Hardware Configuration** screen and select **NSP > Settings > Spike Sorting**.
2. Right-click **Spike Sorting** and select **Properties**.
3. On the **Spike Sorting Properties** dialog box, change the following settings:
 - **Sorting Parameters**
 - **Method** - Select **k-means PCA**.
 - **Wave Basis Size** - Enter the number of waveforms to use to calculate the PCA basis vectors.
 - **Update Basis Multiplier** - The program recalculates PCA basis vectors after the number of samples exceeds the basis size times the basis multiplier you enter. No recalculations will be made if you enter **0** (zero).
 - **Outlier Removal** - Removes outliers exceeding the number of standard deviations from the centroid of each group you enter.
 - **Minimum Group Size** - Removes any groups that have fewer than the number of spikes you enter.

-
- **Merge Centroids** - Merges two groups if the merged group has a standard deviation less than the multiplier you enter times the product of the standard deviations of both groups.
 - **Number k-means groups** - Enter the number of k-means groups to use for sorting.
 - Artifact Rejection. For more information, see **Artifact Rejection** on page 63
 - Click **OK** to apply the settings and close the dialog box.
4. Go to the **Central** main screen and select **Tools > Auto Spike Sorting > K-means PCA** to sort the channels.

Note: Ensure the channels of interest have been configured to use the k-means PCA sorting algorithm. See **To configure the hardware settings for one or more analog inputs:** on page 37.
 5. Go to the **Single Neural Channel Configuration** screen and select the channel of interest on the top menu bar. For example, **chan1**.

Note: If the **Online Classification** pane is hidden, select the **View Classification** button on the top menu bar.
 6. Select the **Feature** tab and then select **Build PCA**.

Note: A counter at the lower left of the screen shows how many of the total required waveforms have been collected. By default, the program needs 250 waveforms to build the PCA basis vectors. The time required to record 250 waveforms will depend on the spike rates and may take several minutes. To modify the number of waveforms, go to **Hardware Configuration > Spike Sorting**. After the feature points have been generated, clustering accuracy will improve as the number of feature points increases.
 7. Select **K-means** to define separate groups.

Note:

 - To save k-means PCA sorting rules, go the **Central** main screen and select **File > Save System Settings**.
 - To load k-means PCA sorting rules, go the **Central** main screen and select **File > Load System Settings**.
 - To save PCA basis vectors, go the **Central** main screen and select **File > Save PCA Basis**.
 - To load PCA basis vectors, go the **Central** main screen and select **File > Load PCA Basis**.

Density-Based Spatial Clustering of Application with Noise (DBSCAN) PCA

Density-Based Spatial Clustering of Application with Noise (DBSCAN) is a principal component analysis method that automatically chooses the quantity, radius, and position of each cluster based on point density.

Figure 9-36: Spike Sorting Properties - DBSCAN PCA

Spike Sorting Properties

Sorting Parameters

Method: DBSCAN PCA

Wave Basis Size: 250 The number of waveforms to use to calculate the PCA basis vectors

Update Basis Multiplier: 0 This number is used to recreate the basis after so many waveforms

Artifact Rejection

Maximum Simultaneous Channels: 272 If X channels fire at the same time their events are rejected.

Refractory Period (samples): 30 Spike detection is turned off for X samples after an event has been detected.

OK Apply Cancel

To set up DBSCAN PCA:

1. Go to the **Hardware Configuration** screen and select **NSP > Settings > Spike Sorting**.
2. Right-click **Spike Sorting** and select **Properties**.
3. On the **Spike Sorting Properties** dialog box, change the following settings:
 - **Sorting Parameters**
 - **Method** - Select **DBSCAN PCA**.
 - **Wave Basis Size** - Enter the number of waveforms to use to calculate the PCA basis vectors.
 - **Update Basis Multiplier** - The program recalculates PCA basis vectors after the number of samples exceeds the basis size times the basis multiplier you enter. No recalculations will be made if you enter **0** (zero).
 - **Artifact Rejection**. For more information, see **Artifact Rejection** on page 63
4. Go to the **Central** main screen and select **Tools > Auto Spike Sorting Method > DBSCAN PCA** to sort the channels.

*Note: Recording additional spike events helps to yield accurate unit definitions. Ensure the channels of interest have been configured to use the DBSCAN PCA sorting algorithm. See **To configure the hardware settings for one or more analog inputs:** on page 37.*
5. Go to the **Single Neural Channel Configuration** screen and select the channel of interest on the top menu bar.

*Note: If the **Online Classification** pane is hidden, select the **View Classification** button on the top menu bar.*
6. Select the **Feature** tab and then select **Build PCA**.

Note: A counter at the lower left of the screen shows how many of the total required waveforms have been collected. By default, the program needs 250 waveforms to build the PCA basis vectors. The time required to record 250 waveforms will depend on the spike rates and may take several minutes. To modify the Wave Basis Size (number of waveforms), go to **Hardware Configuration > Spike Sorting** as described in **Manual PCA**, in the note on page 58. After the feature points have been generated, clustering accuracy will improve as the number of feature points increases.

7. Select **DBSCAN**.

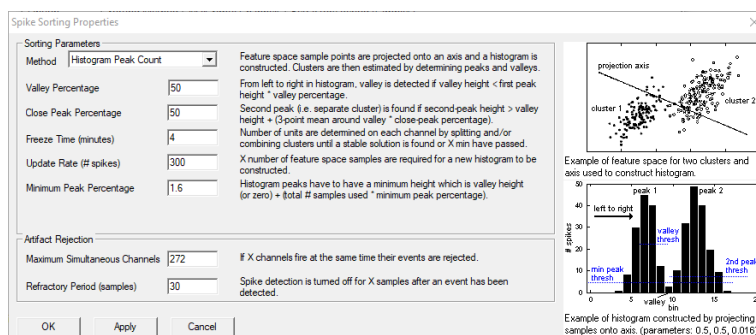
Note:

- To save DBSCAN PCA sorting rules, go the **Central** main screen and select **File > Save System Settings**.
- To load DBSCAN PCA sorting rules, go the **Central** main screen and select **File > Load System Settings**.
- To save PCA basis vectors, go the **Central** main screen and select **File > Save PCA Basis**.
- To load PCA basis vectors, go the **Central** main screen and select **File > Load PCA Basis**.

Histogram peak count

Collected waveforms are broken down into high frequency and low frequency components. Each spike event is plotted on an axis composed of these two components. A histogram is constructed, and clusters are estimated by determining peaks and valleys.

Figure 9-37: Spike Sorting Properties - Histogram Peak Count



- **Valley Percentage** - A valley is identified at a height less than the first peak height times the valley percentage value.
- **Close Peak Percentage** - The second peak is identified at the height of the valley plus the height of the three-point mean centered at the valley multiplied by the close peak percentage.
- **Freeze Time (minutes)** - Time spent analyzing the number of units on each channel.
- **Update Rate (# spikes)** - The number of spikes that must be collected within each unit before the histogram is updated.

- **Minimum Peak Percentage** - Minimum peak height is calculated by adding the valley height to the total number of samples used times the minimum peak percentage.

Artifact Rejection

Figure 9-38: Spike Sorting Properties- Artifact Rejection

- **Maximum Simultaneous Channels** - Enter the maximum number of channels that may contain simultaneous spike events before the events are all rejected.
- **Refractory Period** - Enter the number of samples that will not undergo spike detection following the detection of a spike.

Spike Sorting Properties

Sorting Parameters

Method: **Manual PCA**

Wave Basis Size: **250** The number of waveforms to use to calculate the PCA basis vectors

Update Basis Multiplier: **0** This number is used to recreate the basis after so many waveforms

Artifact Rejection

Maximum Simultaneous Channels: **272** If X channels fire at the same time their events are rejected.

Refractory Period (samples): **30** Spike detection is turned off for X samples after an event has been detected.

OK Apply Cancel

9.2.4.7 Adaptive filtering

Recorded signals might contain unwanted information from sources of external noise, including magnetic eye-tracking systems, experimental monitors, subject reward systems, and equipment racks. Adaptive filtering uses a noise signal as a reference to remove a known noise signal from neural data.

Figure 9-39: Adaptive Filtering dialog box

To use adaptive filtering:

1. Connect the peripheral noise source to an analog input channel on the NSP using an analog input BNC connector.
2. Go to **Hardware Configuration > Adaptive Filtering**.
3. Right-click the channel of interest and select **Properties**.
4. On the **Adaptive Filtering** dialog box, select an operation type (**2 Chan Spike + Continuous Data**, for example).
5. Under **Reference Channels**, select the analog inputs that will be used for the noise sources.

Adaptive Filtering

2 Chan Spike + Continuous Data

Reference Channels

ainp1 ainp2

OK Cancel Apply

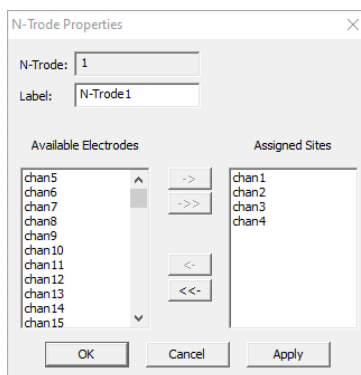
9.2.4.8 N-Trode groups

N-Trode channel groupings are user-defined electrode subsets that can be created to record signals from proximal regions. When any single unit spike is detected within the N-Trode, the signals from all component electrodes are saved and registered as events. Central supports multiple N-Trode sets containing up to four electrodes each. Multiple single channels are linked, and spike sorting occurs on the N-Trode as a single entity.

To define N-Trode groups:

1. On the **Hardware Configuration** dialog box, select **NSP > Settings > N-Trode Groups**.
2. Right-click the N-Trode of interest and select **Properties**.
3. Use the **N-Trode** dialog box to assign available electrodes to the N-Trode:
 - To assign an electrode, select an option (**chan1**, for example) and click **→**, or double-click an option (**chan3**, for example).
 - To assign all available electrodes, click **→>**.
 - Assigned electrodes are listed under **Assigned Sites**.

Figure 9-40: N-Trode Properties dialog box



9.2.5 Renaming Multiple Channels

Central has a built-in, automatic channel naming scheme with several parameters.

1. On the **Hardware Configuration** screen, press **Ctrl** while selecting a range of non-contiguous channels, or press **Shift** while selecting a range of contiguous channels.
2. Right-click the selection and select **Properties**, or click the **Properties** icon on the toolbar.
3. Click **Edit**.
4. Enter the label prefix you want to use across the channels you selected.
5. Choose the numbering scheme you want to apply to each file after the prefix as follows:

- **Use current numbering** - Select this option to maintain the current channel numbering.
- **Number channels sequentially starting at** - Select this option and enter a number in the box to increment the numbering from the start value by adding one to each additional channel.
- **Number using channel ID** - Select this option to number with the channel ID value (1 to 128).
- **Use terminal numbering** - Select this option to number with the pin and bank value of the channel location on the front-end amplifier.

Figure 9-41: Label Options

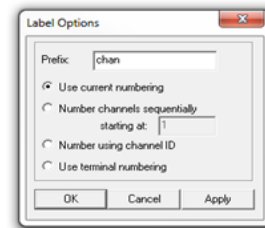


Figure 9-42: Properties icon



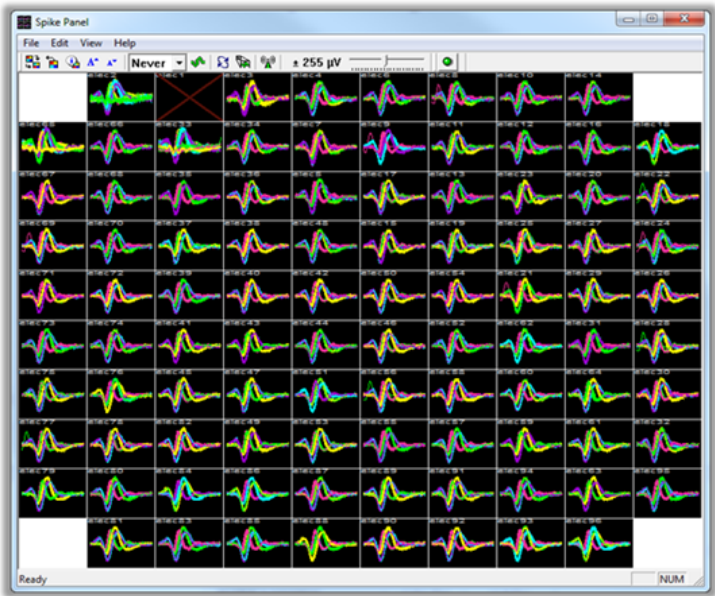
9.3 Spike panel

The **Spike Panel** screen displays detected spikes on the front-end amplifier channels in a grid-graph form, and provides you with an easily accessible overview of the system. Double-click any channel to display the **Single Neural Channel** dialog box.

Right-click anywhere on the screen and select **Options** to change the layout and graphical settings of the grid and individual channels.

Apply a map file to the display to change the spatial arrangement of the channels on the screen to match the electrode array in use. If spike-sorting rules are defined, the spikes belonging to different units will display in different colors.

Figure 9-43: Spike Panel dialog box



9.3.1 Spike panel settings

Figure 9-44: Spike Panel screen

1. **Load Map File** - Select this option and use the dialog box that opens to load custom map files matching the spatial arrangement of your electrode ship with each Blackrock Electrode Array. To request an additional copy of any Utah Array map files, contact Blackrock Support at support@blackrockneuro.com.
2. **Load Default Map File** - Select this option to load the default channel map. The default map file for the spike panel is a 16 x 8 grid.
3. **Mapfile Information** - Display information about the current map file.
4. **Make Title Text Larger** - Enlarge the text size of the title.
5. **Make Title Text Smaller** - Reduce the text size of the title.
6. **Time to Screen Refresh** - Set the length of time spike waveforms will trace without refreshing.
7. **Open Selected Channels** - Individual or groups of channels can be viewed by selecting Open Selected Channels in the right-click menu or the top menu bar.
8. **Refresh Screen** - Manually refresh the spike panel display.
9. **Magnify Current Channel** - Toggle mouse-hover magnification.
10. **Show Popup Labels** - The screen displays a label when you hover over a channel.
11. **Broadcast Channel Selections** - Duplicates channel selections on other instances of Central running on computers connected to an Ethernet switch.
12. **Spike Scale** - Move the slider to change the visual amplitude of displayed spikes.
13. **Level of Criticality** - Icon represents the current application load. Click the icon to view a list of the levels of criticality (see the figure below).

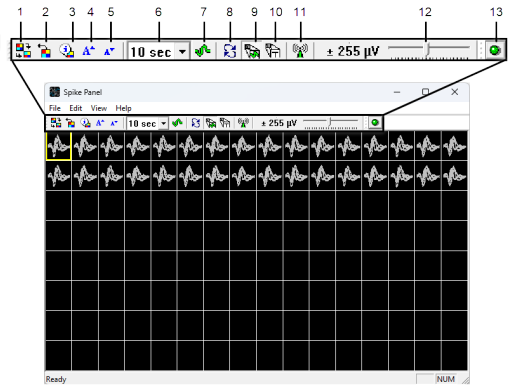
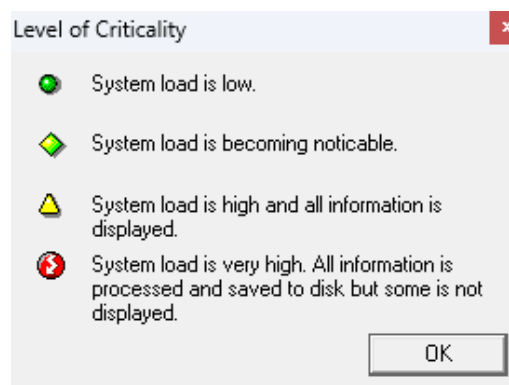


Figure 9-45: Level of Criticality

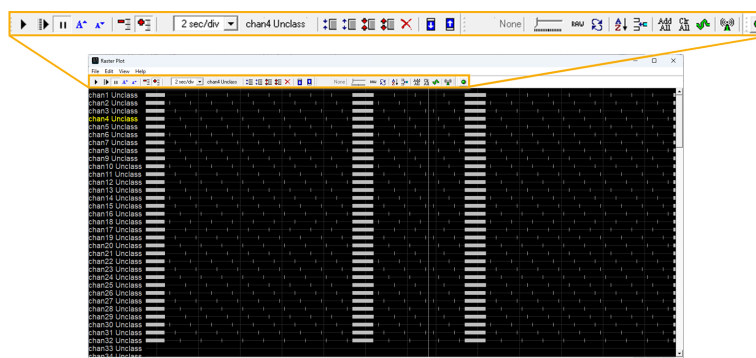


9.4 Raster plot

The **Raster Plot** screen displays the occurrence of spike events on many channels over time. Use this screen to add, remove, move, and scale all input channel types.

The plot also displays continuous traces and comments.

Figure 9-46: Raster Plot screen tool bar



The plot also displays continuous traces and comments.

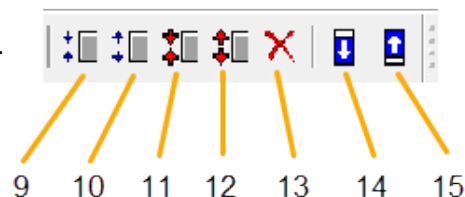
Figure 9-47: Raster Plot screen tool bar - labels 1-8

1. **Play Data** - Configure the display to fill and then refill repeatedly.
2. **Scroll** - Configure the display to scroll right to left.
3. **Pause** - Pause the display.
4. **Increase Font Size** - Resize the text on channel labels.
5. **Decrease Font Size** - Resize the text on channel labels.
6. **Show One Line per Channel** - Display all units from each channel on one line.
7. **Split Channels in Multiple Lines** - Display each unit on a separate line.
8. **Seconds per Division** - Change the time scale markers on the plot.



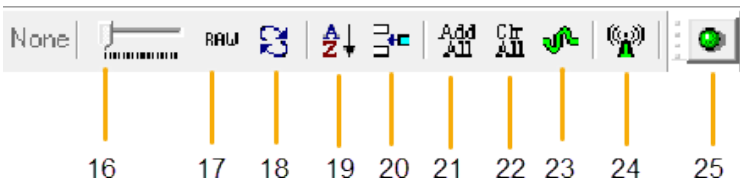
Figure 9-48: Raster Plot screen tool bar - labels 9-15

9. **Decrease Size** - Change the size of selected channels.
10. **Increase Size** - Change the size of selected channels.
11. **Minimize** - Minimize the size of selected channels.
12. **Maximize** - Maximize the size of selected channels.
13. **Delete** - Remove selected channel from the plot.



14. **Move Down** - Move one or more channels you selected down the channel list.
15. **Move Up** - Move one or more selected channels you selected up the channel list.

Figure 9-49: Raster Plot screen tool bar - labels 16-25

- 
16. **Scale Continuous Data** - Shows the amplitude setting of continuous signals. To change the setting, move the slider.
 17. **Toggle Between RAW and Continuous Analog Data** - Switch between filtered continuous analog data and raw data.
 18. **Update Display** - Refresh the display.
 19. **Sort** - Sort channels in the raster plot display by Channel ID, Recording Mode, or Unit ID.
 20. **Choose Channels/Units** - Select channels to display.
 21. **Add All** - Display all channels.
 22. **Clear All** - Remove all channels from display.
 23. **Open Selected Channels** - Open selected channels in Single Neural Channel.
 24. **Broadcast Channel Selections** - Duplicate channel selection on other instances of Central running on computers connected to the Ethernet switch.
 25. **Level of Criticality** - Display the current application load.

9.4.1 Configure raster plot

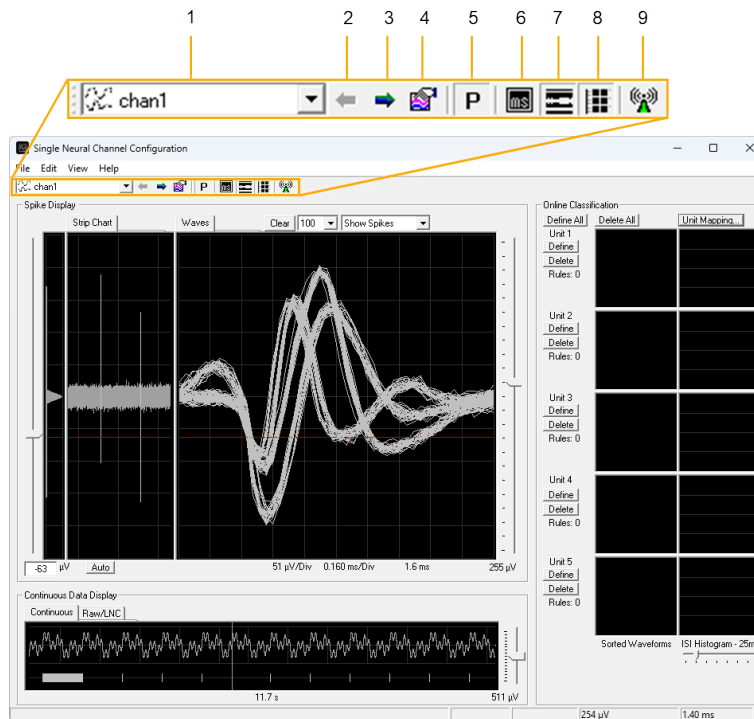
The Raster Plot can display the occurrence of spike events or continuous signals for many channels over time. The plot can be configured to match your experimental setup.

1. Begin by removing all channels from the display using **Clear All**.
2. Open **Choose Channels** and select the channels of interest.
3. Use the **Move** arrows to order the selected channels.
4. Click **Split Channels** to split all channels into component channels. This will display a line for each classified spike unit as well as a continuous signal trace
5. Click **Delete** any unwanted channel components using Delete.
6. Click **Resize** to change the size of the channels of interest. It is often helpful to increase the size of continuous data traces.
7. Adjust the **Seconds per Division** of the markers on the plot.
8. Click **Increase / Decrease Font Size** to adjust the channel labels font size to be easily read.

9.5 Single Neural Channel screen

Use the **Single Neural Channel** screen to visualize the data from a channel on various charts, manage units mappings and definitions, and configure hardware configurations.

Figure 9-50: Single Neural Channel Configuration screen - toolbar



1. **Channel ID** - Select a channel of interest in this list.
2. **Back** - Go to the previous channel.
3. **Next** - Go to the next channel.
4. **Channel Properties** - Open the Channel Properties screen.
5. **Persist Spikes** - View the last detected spikes in the wave viewer for sorting.
6. **View Tool Tip** - Display time and amplitude values below the mouse pointer.
7. **View Continuous** - Select this toggle to show or hide the Continuous Data Display pane.
8. **View Classification** - Select this toggle to show or hide the Online Classification pane.
9. **Send Channel Selections to PCs on Instrument Network** - Duplicate the channel selection on other instances of Central running on computers connected to the Ethernet switch.

9.5.1 Visualizing a channel

Select a channel of interest in **Channel ID** to visualize its data on the **Single Neural Channel** screen:

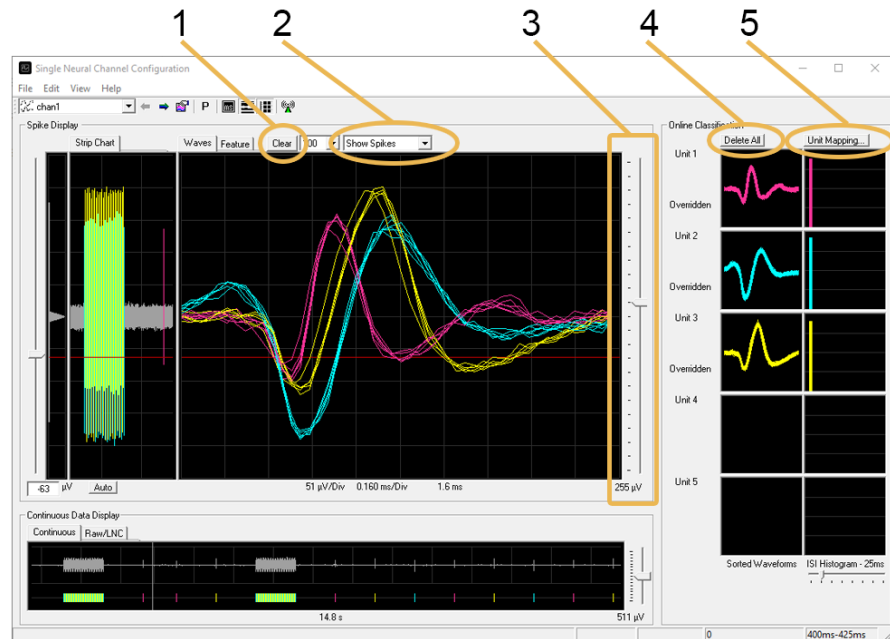
- **Spike Display**- This pane provides two charts:
 - **Strip Chart** - The channel's continuous signal with high-pass filtering applied to the data.
 - **Waves** - A waveform view of the channel.
- **Continuous Data Display** - View the **Continuous** or **Raw / LNC** chart in this pane. Select the **View Continuous** toggle on the toolbar to hide or show this pane.

9.5.2 Spike Display pane

The **Single Neural Channel Configuration** screen displays neural events. It contains the wave panel, which shows spikes, defines sorting hoops, and provides the strip chart.

Figure 9-51: Spike controls on the Single Neural Channel Configuration screen

1. **Clear** - Select this option to delete the cached samples and clear the spikes displayed in the waves pane. Central refreshes the samples and plots the spike units you selected in **Show Spikes**.
2. **Show Spikes** - Select an option in this list to filter the spike units in the waves pane.
3. **Voltage slider** - Move the vertical slider to change the voltage range. (The default time scale is 1.6 milliseconds.)

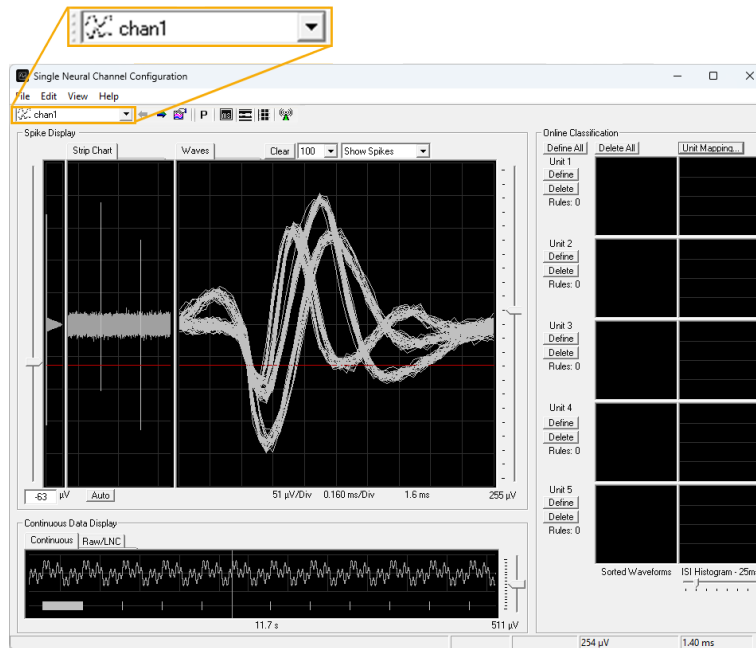


- Note: Use the **Global Settings** dialog box to set the sample width for each spike between 30 and 128 (1.0 to 4.3 milliseconds). See **9.2.4.1 Global settings** on page 50.
4. **Define All** - Select this option to display the color-coded selection handles, and then drag a handle to each waveform you want to assign to a unit.
5. **Unit Mapping** - Select this option and use the dialog box that opens to select the unit definitions you want to remap, combine, or map to noise.
6. **Define** - Each of the 5 **Unit** sections on the **Online Classification** pane provides a **Define** option. Select this option to display a color-coded selection handle and then drag a handle to the waveform you want to assign to the unit. A color-coded selection handle will appear at the lower right of the **Spike Display** pane.

9.5.2.1 Waves chart

Select a channel of interest in **Channel ID** to visualize its data on the **Single Neural Channel Configuration** screen.

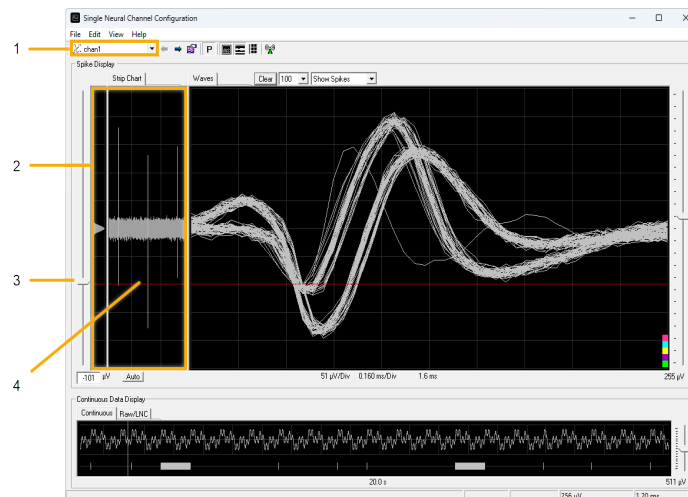
Figure 9-52: Channel ID selection on the Single Neural Channel Configuration screen



9.5.2.2 Strip chart

The **Strip chart** visualizes the selected channel's continuous signal with high-pass filtering applied to the data.

Figure 9-53: Single Neural Channel Configuration screen - Strip Chart pane



1. **Channel ID**- Select a channel in this list to visualize its data.
2. **Strip chart** - This charts the continuous signal with high-pass filtering applied to the data
3. **Threshold** - Move the slider up to increase the threshold or down to do the opposite.
4. **Red line** - Marks the threshold for filtering the spikes.

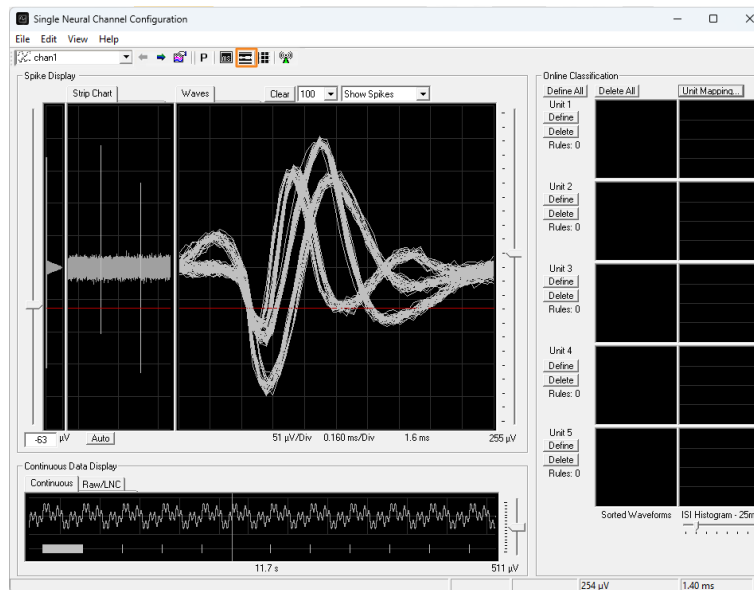
9.5.3 Continuous Data Display pane

Select **View Continuous** on the toolbar of the Single Neural Channel screen to display this pane.

To switch between displays:

- **Continuous** - Select this option to display a continuous data trace and a raster plot. The ticks below the raster plot show the relationship of waveforms to the continuous data.
- **Raw / LNC**.- Select this option to display a constant subtracted waveform generated by line noise cancellation.

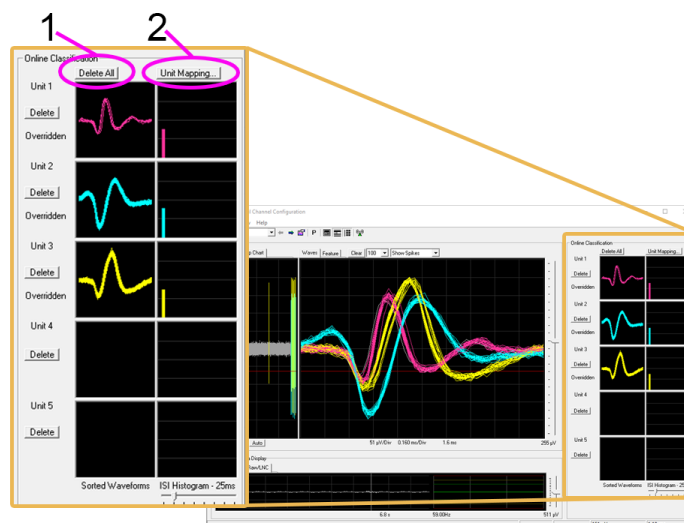
Note: This option must be enabled.

Figure 9-54: View Continuous on the Single Neural Channel Configuration screen

9.5.4 Online Classification pane

Use the **Online Classification** pane to map units and manage unit definitions.

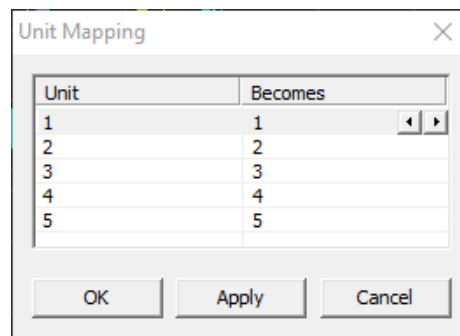
To display this pane, select the **View Classification** toggle on the toolbar to hide or show this pane.

Figure 9-55: Online Classification pane

1. **Unit Mapping** - Select this option and use the dialog box that opens to select the unit definitions you want to remap, combine, or map to noise (see the image below).

2. **Delete All** - Deletes all of the spike sorting for the selected channel.
3. **UnDefine / Define All**- (not shown) Select the **Define All** toggle to display the color-coded selection handles you use to assign waveforms to units. The histogram block on the side shows the Inter-Stimulus Interval (ISI) histogram for each of the single units. Select **UnDefine** to undo the assignment.
4. **Define** - (not shown) Each of the 5 **Unit** sections on the **Online Classification** pane provides a **Define** option. Select this option to display a color-coded selection handle and then drag a handle to the waveform you want to assign to the unit. A color-coded selection handle will appear at the lower right of the **Spike Display** pane.

Figure 9-56: Unit Mapping dialog box

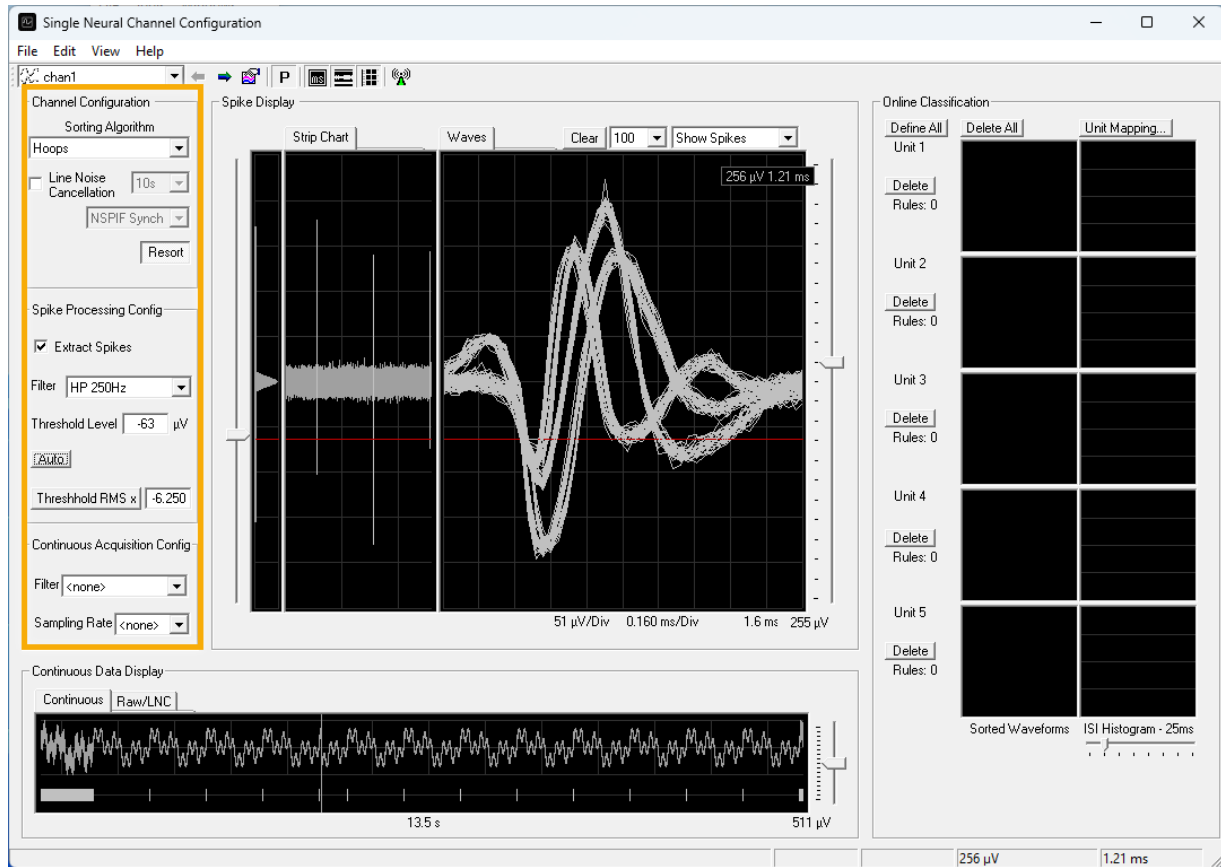


9.5.5 Configuration pane (optional)

The optional **Configuration** pane provides convenient access to the **Channel Configuration**, **Spike Processing Config**, and **Continuous Acquisition Config** hardware settings on the **Single Neural Channel Configuration** screen. (These settings also appear on the **Hardware Configuration** screen.)

To display the **Configuration** pane:

1. Go to the **Central** screen and select **Tools > Options**.
2. Select **Filter Rate Separate** and **Configuration on Window** on the **Windows Options** dialog box.
3. Click **OK** to apply the settings.

Figure 9-57: Single Neural Channel Configuration screen - Configuration pane

9.6 File storage

Central supports file storage using the following interfaces:

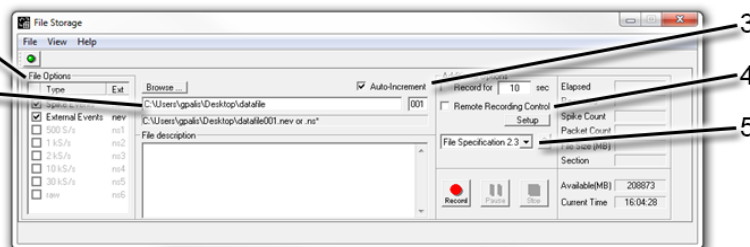
- **Research Interface** - The most common basic file interface for research.
- **TOC** - Names files with a time stamp, includes patient information, and allows long recordings to be split into multiple files. Data files are saved in NSX format. NSX data is stored in bits with 2, 0 nV/bit resolution. Timestamps appended to file names by Central use Greenwich Mean Time (GMT) referencing your local computer's clock and time zone information.

9.6.1 Research interface

The Research Interface allows you to specify a file name and path for the next data set for research settings. Use this window to start or stop recording or to stop recording after a specified duration.

Figure 9-58: Research Interface

1. **File Options** - Select the sampling frequency for the recorded data.
2. **File Name and Path** - Select or enter the desired file name and path.
3. **Auto-Increment** - Select this option to automatically increment the data file names with an appended numeral.
4. **Remote Recording Control** - Allows digital in signals to control file recording through the digital input port or the serial I/O port. Click the **Setup** button below the remote recording check box to set user-defined hex values for each remote recording function. For more information, see **9.6.4 Remotely control data recording** on page 80.
5. **File Specification** - Select the applicable file specifications. (Always use the most current specification.)



9.6.2 TOC interface

The TOC interface allows the user to enter patient information and select a path for the files that are created when the session is initiated. Central assigns a numeric name to the files which indicates the date and time of the recording. For example, *20240124-152145* indicates that the data file was recorded on January 24, 2024, at 3:21:45 PM GMT. System settings will be saved in the same folder as the data file.

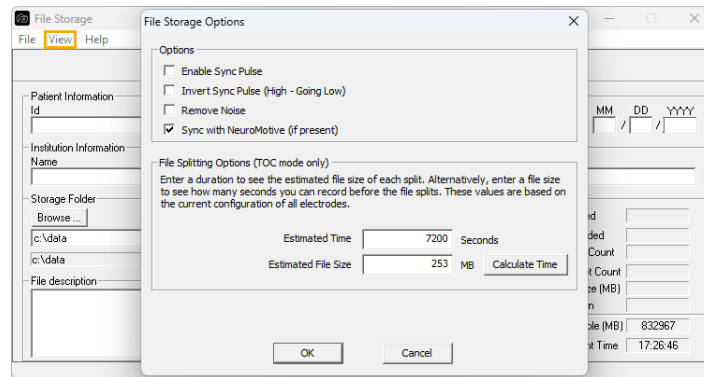
Figure 9-59: TOC Interface

1. **File Description** - Up to 256-character comments can be entered here to be saved with the file. Comments need to be entered prior to recording a data file.
2. **Record For** - Specify an amount of time that Central will record data. Once the time has expired, the recording will stop automatically.

-
3. **Remote Recording Control** - Allows digital in signals to control file recording through the digital input port or the serial I/O port. Click the **Setup** button below the remote recording check box to set user-defined hex values for each remote recording function. For more information, see **9.6.4 Remotely control data recording** on page 80.
 4. **Disable File Splitting** - Select to disable file splitting in the file storage module. Splitting large files into smaller sizes may aid offline analysis. File size limit is defined in the **View > Options** menu.
 5. **Record** - Begin recording data.
 6. **Stop** - Stop recording and save file.

9.6.3 File storage options

Figure 9-60: File Storage window



To update the File Storage Options settings under **Options**, go to the **File Storage** screen and select **View**:

- **Options**
 - **Enable Sync Pulse** - Send a unique waveform pattern every 14 seconds and repeat patterns hourly.
 - **Invert Sync Pulse** - Invert the high and low phases of the sync pulse.
 - **Remove Noise** - Do not save waveforms classified as noise during spike sorting.
 - **Sync with NeuroMotive** - Enable syncing with the NeuroMotive video recording and tracking system.
- **File Splitting Options**
 - **Split recordings by duration** - Split recordings into segments no longer than your entry in **Estimated Time**, and refer to **Estimated File Size** for the estimated file size of each resulting segment.
 - **Split recordings by file size** - Split file recordings into segments no larger than the file size you enter in **Estimated File Size**, and refer to **Estimated Time** for the estimated duration of each resulting segment.
- Click **OK** to apply the updates and close the dialog box.

9.6.4 Remotely control data recording

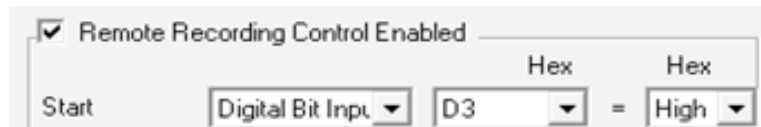
The Central software can be configured to start and stop recording when triggered by a digital input coming from other hardware such as a behavioral control system. When using the TOC file format, the recording can be remotely started, stopped, resumed, or a new session can be initiated. When using the Research Interface file format, the recording can be remotely started, stopped, paused, and resumed.

To set up remote recording using the DB-9 serial port or DB-37 digital input port:

1. Select **Remote Recording Control Enabled**.
2. Click **Setup**.
3. In the **Start** menu, select **Digital Bit Input** for any of the four remotely controllable tasks.
4. In the first **Hex** menu, choose a hex bit for this function.
5. In the second **Hex** menu, choose **High** or **Low** for triggering.

For example, to begin recording remotely when digital bit D3 on the digital input port is set high, configure as shown below.

Figure 9-61: Example Settings for Remote Recording



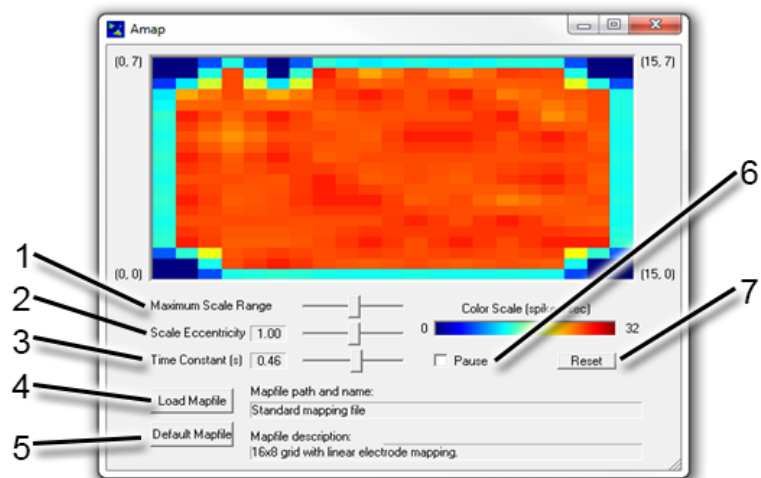
9.7 Activity map

Use the activity map (**Amap**) to color-code the firing rate of neurons on a two-dimensional map. An electrode map file assigns each channel to a location. The spiking activity of each channel is binned and smoothed over a fixed time window of 20 milliseconds.

Smoothing is a technique that applies a mathematical function to the data of an activity map eliminates noise and inconsistencies. Smoothing reveals underlying patterns that are not easily detectable in the original data can be made more visible. Activity map uses Gaussian smoothing, which uses a filter that follows the shape of Gaussian distribution to reduce high frequency noise. The technique also adds steps between events and non-events between channels and over time.

Figure 9-62: Amap dialog box

1. **Maximum Scale Range** - Set the maximum value of the color scale in spikes per second.
2. **Scale Eccentricity** - Skew the mapping of spike rate to color towards high or low values.
3. **Time Constant** - Set the time constant controlling activity decay.
4. **Load Map File** - Load a map file to change the orientation of the display.
5. **Default Map File** - Return to the default 16 x 8 grid map file.
6. **Pause** - Pause the display until unchecked.
7. **Reset** - Clear current activity data and refresh visualization.



9.7.1 Change electrode map (create a map file)

The electrode positions on **Spike Panel** and **Activity Map** can be rearranged. A plain text “map” file specifies the location assignments of each channel.

Figure 9-63: Sample CMP File

```
// 8-Channel NSP map file demo
//
// Data is as follows, 'C(olumn) r(ow) b(ank) e(lectrode) l(abel)'
// c - 0 based column from left to right
// r - 0 based row from bottom up
// b - bank name - values can be A, B, C, or D
// l - label used to rename channels in Central
//
// Comments begin with '//'
// First non-comment line is the Mapfile description
//8 Channel NSP mapping
0          1          A          4          elec08
1          0          A          8          elec07
2          1          A          3          elec06
3          0          A          6          elec05
0          0          A          5          elec01
1          1          A          2          elec02
2          0          A          7          elec03
3          1          A          1          elec04
```

The map file's columns provide the following information:

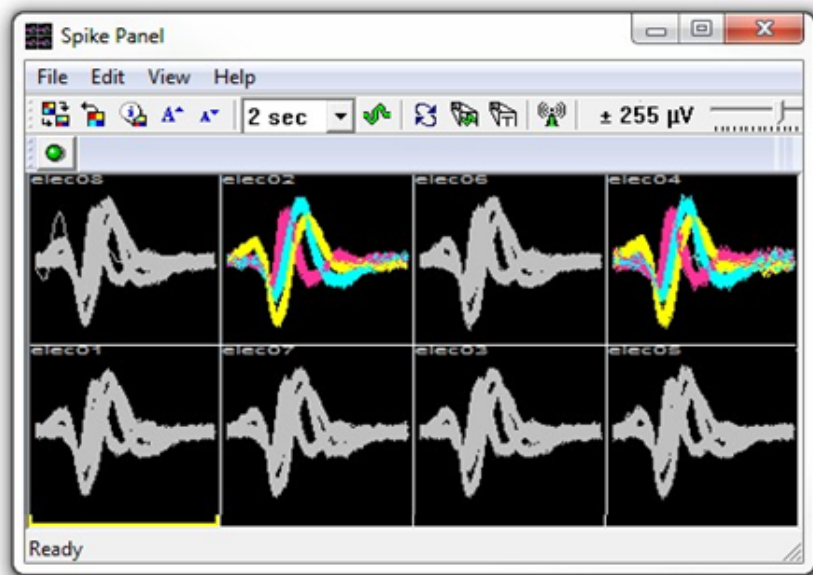
- **Column 1** - Location of the spike window on the x-axis. (Location positions start from 0.)
- **Column 2** - Location of the spike panel on the y-axis. (Location positions start from 0, and point 0,0 is at the lower left corner of the screen.)
- **Column 3** - The bank on which the physical channel is located (A, B, C, or D).
- **Column 4** - The bank pin to which the channel is connected.
- **Column 5** - An optional label for the channel.

The preceding example shows:

- The first line is `0 1 A 4 elec08` which indicates the channel on bank A pin 4 is named "elec08". This channel is located in the first column from the left (0) and on the second row from the bottom (1).
- The last line is `3 1 A 1 elec04` which indicates the channel on bank A pin 1 (channel 1) is named "elec04". This channel is located in the fourth column (3) from the left and on the second row from the bottom (1).

- Comments begin with two slashes (//) and they are ignored, such as
// c - 0 based column from left to right.
- The line following the last set of slashes names the map file (as referenced in the apps) and it is also ignored:
//8 Channel NSP mapping.

Figure 9-64: Sample Spike Panel referenced in the map file

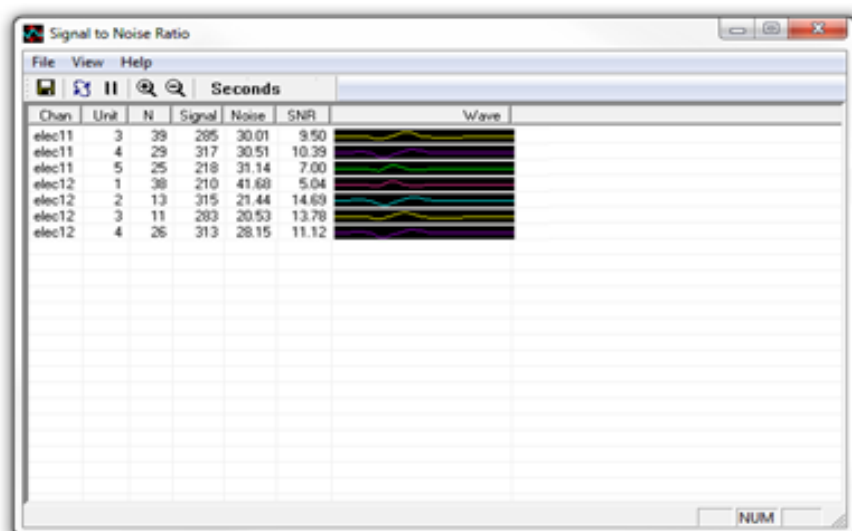


9.8 Signal-to-noise ratio

Use the **Signal to Noise Ratio (SNR)** screen to view the running amplitudes of the signal, noise, and signal-to-noise ratio for units on each channel. This screen displays the signal-to-noise ratio as a function of time, which allows you to monitor changes in signal quality. You can save this information as a tab-delimited text file, and open it in Excel or a compatible program for further analysis.

In the example below, the SNR for **elec11 unit 3** is **9.50**, which means that the signal is 9.5 times greater than the noise.

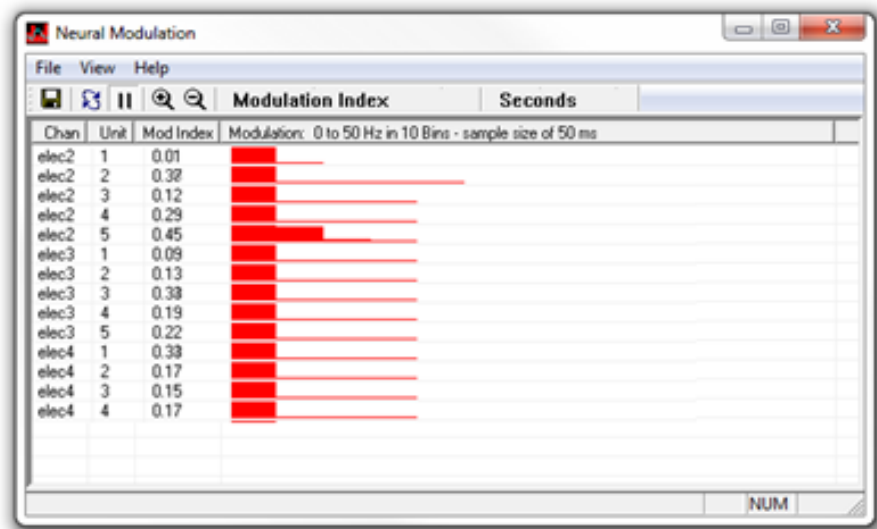
Figure 9-65: Signal to Noise Ratio screen



9.9 Neural modulation

Use the **Neural Modulation** screen to view a histogram of changes in the firing frequency over time. The **Modulation Index** is the variance of the histogram entries, which covers a range of 0 to 10. A range of 2 to 5 is considered normal when recording neural signals.

Figure 9-66: Neural Modulation screen



9.10 Impedance tester

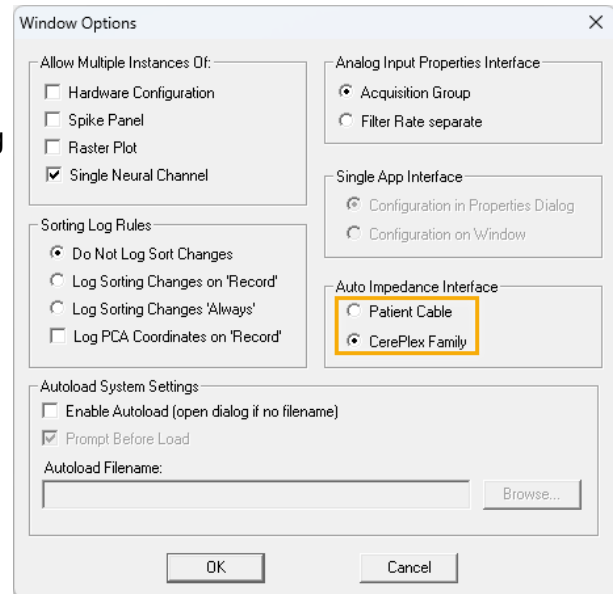
The Impedance Tester measures and reports electrode impedance on each channel.

Note: Special equipment may be required for impedance measurements. Contact Blackrock Microsystems sales at sales@blackrockneuro.com for more information.

Figure 9-67: Windows Options dialog box - Auto Impedance Interface options

You can select a mode for the Impedance Tester based on the connected headstage:

- **Patient Cable** - Select this mode when using an analog Patient Cable
- **CerePlex Family** - Select this mode when using a digital headstage.



To select a mode for the Impedance Tester:

1. On the **Central** main screen, select **Tools > Options > Auto Impedance**.
2. On the **Windows Options** dialog box, select an **Auto Impedance Interface** mode and then click **OK**:
 - **Patient Cable** - Select this mode if an analog Patient Cable is connected.
 - **CerePlex Family** - Select this mode if a digital headstage is connected.

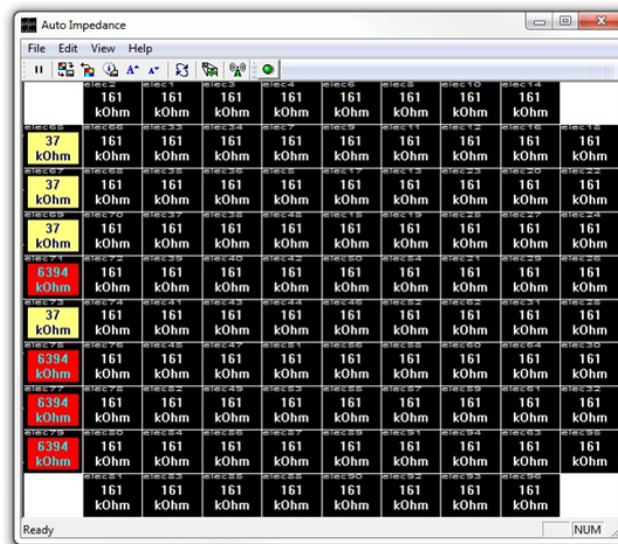
Figure 9-68: Auto Impedance dialog box

To view tabulated information about the electrode impedance on each channel:

1. On the **Central** main screen, select **Impedance Tester**.
2. Complete the **Auto Impedance** dialog box (see the figure above) as follows:
 - **Enter Array S/N** - Select the applicable serial number.
 - **Enter Date of Implantation** - Select the applicable date of device implantation.

- **Type of Sterilization** - Select the applicable type of sterilization.
 - **Comments** - Enter any pertinent comments.
3. Select **Finish> Next**. The **Auto Impedance** screen displays (see the figure below).

Figure 9-69: Auto Impedance screen



9.11 Crosstalk

Use the **Crosstalk** screen to compare each channel and identify cross-correlation. The Crosstalk Diagnostic feature displays a color-coded icon for each channel (green, yellow, or red) indicating the measurement of similarity between the signals detected on each electrode.

Figure 9-70: Crosstalk screen

Color coding for channel icons:

- **Green** - No cross-correlation to any other channel was found.
- **Yellow** - Possible cross-correlation found. Examine these signals closely to determine their best use.
- **Red** - High levels of cross-correlation found. Consider removing these channels from recordings.



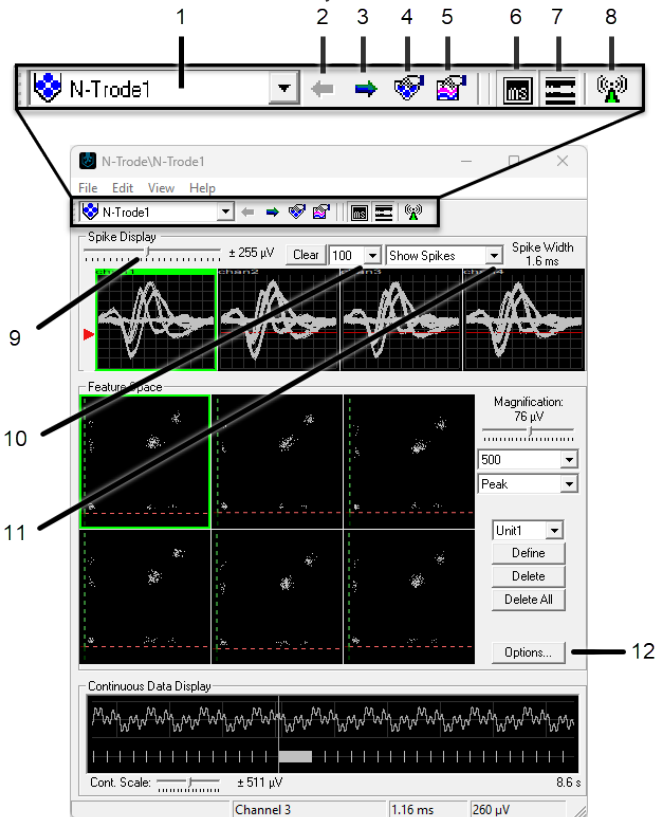
9.12 N-Trode

You can link multiple single channels to N-Trode channel groupings. Groupings are user-defined electrode subsets created to record signals from proximal regions when any single unit spike is detected within the N-Trode. N-Trodes allow for neural events detected on one channel within an N-Trode to force the creation of neural events on the other N-Trode channels. When an event is detected on a channel within an N-Trode, all other N-Trode channels record a neural event with the same time stamp, spike width samples, prethreshold samples, and refractory period. (For more information, see **9.2.4.1 Global settings** on page 50.) Central supports multiple N-Trode sets containing up to four electrodes each. Define N-Trode configurations in the Hardware Configuration window or within the N-Trode application window.

Note: Spike sorting occurs on the N-Trode as a single entity. This causes neural events on all N-Trode channels to be subject to refractory periods caused by a neural event detected on any N-Trode channel.

Figure 9-71: N-Trode\N-Trode1 screen

1. N-Trode - Select an N-Trode group.
2. Next - Click to scroll to the next N-Trode.
3. **Back** - Click to scroll to the previous N-Trode.
4. N-Trode Properties - Define or modify an N-Trode group.
5. Channel Properties - Go to the Analog Input Properties dialog box for all channels in the selected N-Trode group.
6. View Tool Tip - Display time and amplitude values at the cursor position.
7. View / Hide Continuous - Toggle the Continuous Data display at the bottom of the window.
8. Send Channel Selections - Duplicates the channel selection on other instances of Central running on computers connected to the Ethernet switch.
9. Spike Display - Move the slider to adjust the voltage threshold for spike detection on all channels.
10. Clear - Select the number of spikes you want the screen to display, and then click **Clear** to hide the remaining spikes. .
11. Show Spikes - Select which spike units will be displayed.
12. Options- Select to view a list of the keyboard and mouse actions you can use to manipulate the N-Trode GUI. For example, press Z to reset the camera.

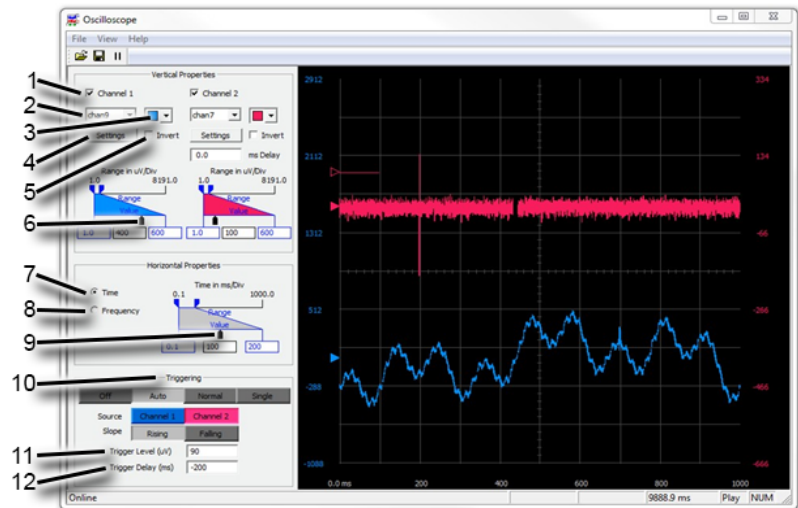


9.13 Oscilloscope

The digital oscilloscope can display and measure two input channels on screen at a resolution of 0.1 milliseconds (ms) per division. Signals can be viewed continuously or on a specified trigger. Frequency spectra can also be displayed.

Figure 9-72: Oscilloscope screen

1. **Channel Enable** - Select to view two channels per instance of the oscilloscope.
2. **Channel Selection** - Choose any channel from the drop-down menu.
3. **Color Selection**: Select a color for either trace.
4. **Settings** - Click to open the **Properties** dialog box for the selected channel.
5. **Invert** - Select to invert the trace for a given channel.
6. **Trace Amplitude** - Change the signal amplitude by moving the sliders for each channel. Move the top slider to select a range. Move the lower slider to select a value within the range you selected.
7. **Time** - Set the oscilloscope to time mode.
8. **Frequency** - Set the oscilloscope to frequency mode.
9. **Time Resolution** - Move the sliders to change the time resolution of the trace. Move the upper sliders to change the range of resolution, and move the lower slider to choose a value within the range you selected.



Note: If no continuous trace is visible on the oscilloscope, confirm that the channel of interest is being sampled from in the Analog Input Properties window. If no trace appears despite sampling from the channel, there may be a compatibility issue with your graphics adapter. Please contact Blackrock at support@blackrockneuro.com.

10. **Triggering** - Different triggers can be used to position and display waveforms on the oscilloscope.
 - **Auto** - A continuous signal will be displayed until the trace crosses the user defined trigger level. When the trigger level is crossed, the trace will pause for two seconds and then resume until another triggering event occurs.
 - **Normal** - The trace will freeze as soon as the trace crosses threshold until the trace crosses the trigger level again, when the screen will update with the new trace.
 - **Single** - A single trace is captured as soon as it crosses the trigger level. To capture another threshold-crossed trace click on **Single** again.
11. **Trigger Level (μV)** - Enter the trigger level in this field or move the trigger line.
12. **Trigger Delay (ms)** - Enter a value to adjust the trigger position on the time axis. At 0 delay, the trigger will appear at the left margin of the screen and might not be visible. A negative value displays the trigger point to the right of its previous position.

9.13.1 Display an input channel on the digital oscilloscope

Set up the digital oscilloscope to view two channels at high resolution. During the setup process, select a trigger to improve the visualization of the signals.

Triggering enables you to capture snapshots of a signal based on its properties. For instance, capturing snapshots of signals that cross a specific threshold voltage. Enabling triggering (depending on the settings) enables you to view points of interest, including spikes or large noise artifacts.

When triggering is disabled, the oscilloscope runs continuously.

To set up the oscilloscope.

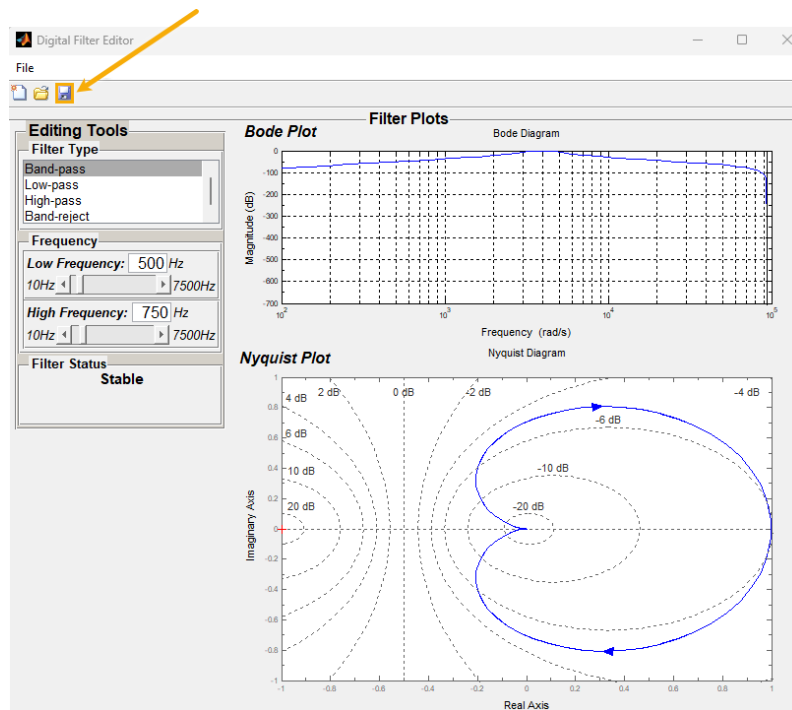
1. Open Central and select **Oscilloscope**.
2. Select one or both options: **Trace 1** and **Trace 2**.
3. Select a channel for each trace on the drop-down menu.
4. Click **Settings** to view and verify all channel settings, including ensuring a sampling rate is selected for the given channel. If **none** is selected for sampling rate, a trace will not appear in the Oscilloscope.
5. Set the amplitude of each trace by moving the slider for each channel. The upper slider modifies the range of values, and the lower slider selects a value within the selected range. The units are $\mu\text{V}/\text{division}$.
6. Move the trace location up or down by dragging the small triangle at the left of the trace.
7. Set the time resolution by moving the resolution slider. The upper slider modifies the range of values, and the lower slider selects a value within the selected range. The highest resolution is 0.1 milliseconds.
8. A trigger can be set for the traces.
 - Choose a channel source for the trigger.
 - Set the trigger level by entering a voltage value in the field or by sliding the trigger line up and down on the screen.
 - Assign a trigger delay. At a default of zero, the trigger will be positioned at the left margin of the screen and may not be visible. Entering negative values reposition the trigger to the right.
9. Select a trigger type of Single, Normal, or Auto.
 - **Auto** - Continuous signal will be displayed until the trace crosses the user defined trigger level. When the trigger level is crossed, the trace will pause for two seconds, then resume until another triggering event occurs.
 - **Normal** - The trace will freeze as soon as the trace crosses threshold until the trace crosses the trigger level again in which case the screen will update.
 - **Single** - A single trace is captured as soon as it crosses the trigger level. To capture another threshold-crossed trace, click **Single** again.

9.14 Digital filter editor

Use the **Digital Filter Editor** screen to configure custom filters for Central. It requires the MATLAB Compiler Runtime (MCR) to be installed. Contact support@blackrockneuro.com for a copy of this software.

Filter type and corner frequencies are defined in the **Editing Tools** section on the left side of the window. The stability of the filter is shown in the **Filter Status** section below. The Bode Plot and the Nyquist Plot of the designed filter are displayed. Once the filter parameters are set, click **Save** (see the icon below) to save the digital filter.

Figure 9-73: Digital Filter Editor screen - Save icon



To use the **Digital Filter Editor** to create custom filters:

1. On the **Central** main screen, click **Digital Filter Editor**.

Note: This tool requires MATLAB Compiler Runtime (MCR), which is available on the Central Suite Installation CD or can be downloaded from www.blackrockneuro.com.

2. On the Digital Filter Editor screen, choose a filter type and corner frequencies.

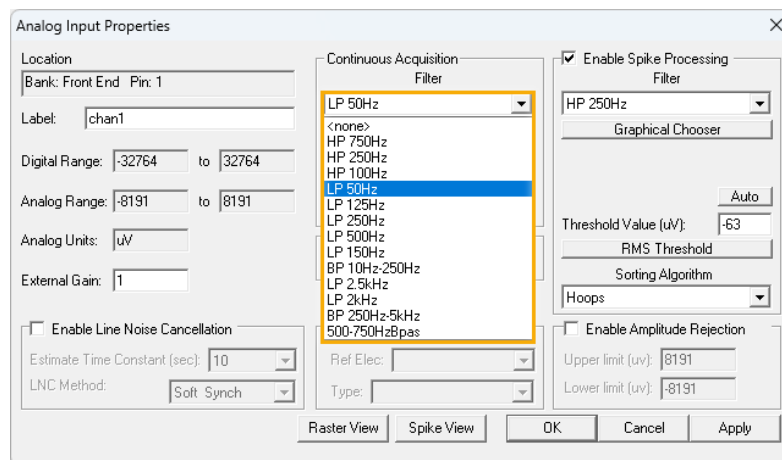
Note: If the filter is unstable, the filter design tool will display a warning and suggest ways to stabilize the filter.

3. Save the filter in a known location and close the Digital Filter Editor screen.

Note: *If you attempt to close the screen without saving the filter, a message asks, "A filter has been changed and not saved. Do you want to close the app without saving your filter changes?" here? Select **No** and save the filter.*

4. Go to the **Central** main screen and select **File > Load Digital Filters**.
5. Use the **Load a Digital Filter** dialog box to select and load up to four custom filters.
6. Confirm the filters were added:
 - Go to the **Hardware Configuration** screen and select **Signal Types > Analog Input** on the tree menu.
 - Right-click a channel and select **Properties**.
 - Expand **Filter** list to view the newly loaded filters.

Figure 9-74: Digital Filter Editor screen - Filter list



9.15 Add comment

The **Add Comment** feature allows the user to add a time-stamped text comment to the Neural Event file. Comments are displayed in the **Raster Plot** view. Use the **Raster Plot** window or the **Central** window to enter a time-stamped comment.

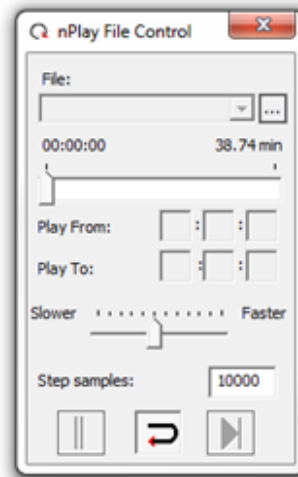
When in the **Raster Plot** view, press any standard letter key on the keyboard to open the **Add Comment** window and start dictating a comment. Press **Enter** when the comment is complete.

9.16 *nPlay* server

Use nPlay Server to load and replay recorded neural data in NSX or NEV formats for viewing and analysis in Central. While replaying the continuous data files, the user can filter, sort spikes, and downsample data.

An NSP does not need to be connected to the computer to use this utility.

Figure 9-75: nPlay Server



10 SETUP

10.1 Hardware setup

10.1.1 Gemini Cerebus system

1. Remove the NSP and Gemini Hub from the shipping boxes.
2. Attach the NSP rack-mount brackets to install it in an equipment rack, or attach the rubber feet to place it on a table.
3. Plug in the NSP and Gemini Hub power cables to electrical outlets.
4. Connect one end of an Ethernet crossover cable to the NSP and the other end to the Gemini Hub.
5. Attach headstages to the Gemini Hub and connect them to electrodes. Make sure the connections are secure.

**WARNING**

ONLY CONNECT HEADSTAGES TO THE GEMINI HUB. NEVER PLUG HEADSTAGES INTO ANY OTHER DEVICE.

6. Turn on the NSP and the Gemini Hub.

Note: Maximum channel capacity setup may include up to four NeuroPlex E headstages, two 256-channel Gemini Hubs, and one NSP.

10.2 Software setup

10.2.1 Installing Central Suite

1. Go to **Control Panel > Programs > Programs and Features > Uninstall a Program** and uninstall previous versions of Central.
2. Run the Central Suite installer (.msi) from the installation CD or blackrockneurotech.com/research/support/#manuals-and-software-downloads.
3. Install the MATLAB Compiler Runtime (MCR) from the supplied installation CD or blackrockneurotech.com/research/support/#manuals-and-software-downloads.

Note: Blackrock Microsystems recommends running Central Suite on a dedicated computer. Do not install additional software or connect the computer to the internet. Disable Windows automatic updates on the computer.

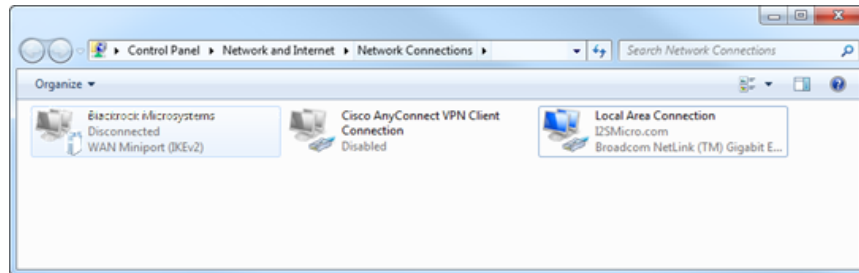
10.2.2 Setting up the Ethernet connection

Note: If you purchased the host computer from Blackrock Microsystems, the computer contains a preinstalled, preconfigured Ethernet adapter.

To connect to the NSP after installing an Ethernet card in the computer:

1. Select **Start > Network Connections**.
2. Right-click the newly installed Ethernet adapter (**Local Area Connection**, for example) and then select **Properties**.

Figure 10-1: Network Connections dialog box



3. Go to **Networking** (see the figure below), select **Internet Protocol (TCP/IP)** or **Internet Protocol Version 4 (TCP/IPv4)**, and then clear the options for all other services.
4. Select **Internet Protocol (TCP/IP)** and select **Properties**.
5. Enter the applicable address in **IP Address**:
 - 192.168.137.1 when connected to legacy / fiber optic Neural Signal Processors.
 - 192.168.137.199 when connected to the Gemini NSP or Gemini Hub.
6. Enter 255.255.255.0 in the **Subnet mask** field.
7. Leave the rest of the fields blank.
8. Select **OK** to save the changes.

Figure 10-2: Local Area Connections Properties dialog box

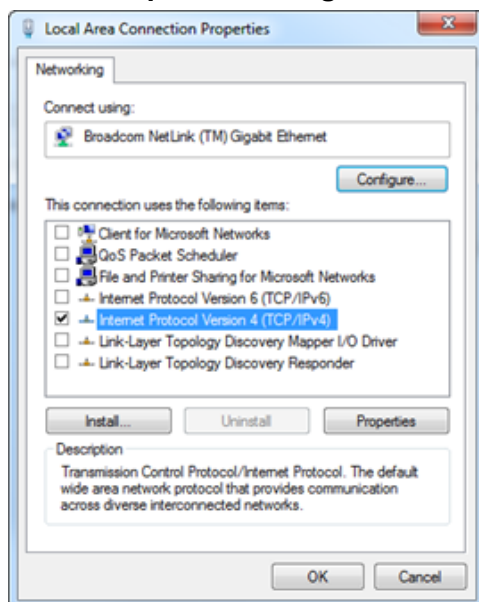
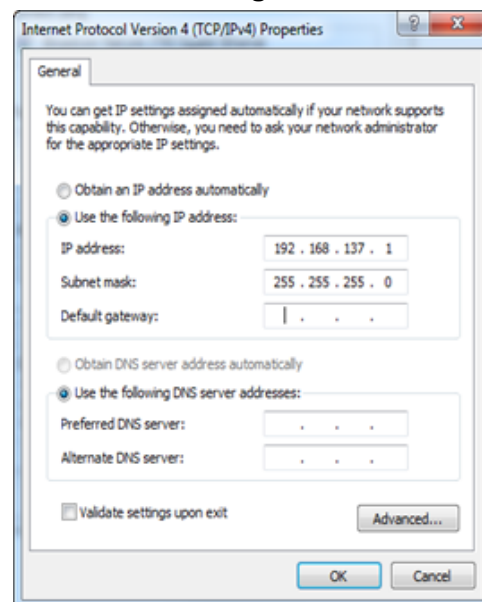


Figure 10-3: IPv4 Properties dialog box



Note: Up to 16 Host PCs can be connected to the same NSP using a business class, 1-Gbps network switch with a QoS feature. To connect more than one Host PC to the network, increment the IP addresses for each additional Host PC. For example, use 192.168.137.2, 192.168.137.3,

The NSP's IP address is 192.168.137.128.

11 TROUBLESHOOTING

Below you can find more information on some of the commonly asked questions.

11.1 Hardware errors

Problem	Symptom(s)	Failure	Potential fix
No data is displayed.	Amplifier / Digital Hub: The Fiber Optic Data Link LED on the front of the Digital Hub or the back of the Amplifier is red or amber. Gemini Hub: The Power LED does not change to green.	The system is not receiving valid data packets.	If the Cerebus system is turned on and running but there is no apparent activity on any of the channels from an implant or from the simulator, verify the following: Ensure that the cables are securely plugged in. Open Central and verify that packets are being received from the Cerebus system. Analog System - Verify that the fiber optic data link between the Amplifier / Digital Hub and the Cerebus NSP is plugged in and that the green light next to the fiber optic data port on the NSP is illuminated. If the light is red or amber, there is no fiber optic link present. Check the LED next to the fiber optic port on the back of the Amplifier to verify that it has power. There may be a compatibility issue or damaged cable. Replace the cable or contact ssupport@blackrockneuro.com with the Digital Hub and headstage serial numbers and the cable lengths. Digital System - Verify that the fiber optic data link between the Amplifier / Digital Hub and the Cerebus NSP is plugged in and that the green light next to the fiber optic data port on the NSP is illuminated. If the light is red or amber, there is no fiber optic link present. Check the LED next to the fiber optic port on the back of the Amplifier or front of the Digital Hub to verify that it has power. There may be a compatibility issue or damaged cable.

Problem	Symptom(s)	Failure	Potential fix
			Replace the cable or contact ssupport@blackrockneuro.com with the Digital Hub and headstage serial numbers and the cable lengths. Gemini System – Verify that the Ethernet crossover cable between the Gemini Hub is connected to the NSP or directly to the host computer.
Noise in the recording.	Poor signal-to-noise ratio or excess high-frequency signal.	Uncontrolled noise sources or poor referencing or grounding scheme.	The system is sensitive to electromagnetic signals. It is designed to detect microvolt level signals. Arrange cables carefully and avoid external noise sources. Avoid allowing cables of the system near other power cables, monitors, light sources, or other electrical equipment. Change which electrode is used as reference to better remove common noise, or use noise cancellation in Central to remove noise from the recording.
Cannot turn on the NSP.	No lights display on the NSP LCD.	Secondary power supply switch is off.	Verify that the Cerebus system is plugged into a working outlet and verify that all power cords are plugged into the NSP. Verify the position of the power switch in the rear of the Neural Signal Processor is ON. If the power switch is in the ON position, turn it to the off position then wait at least five seconds before turning the NSP back on. (When restarting the NSP, always wait at least five seconds before turning the unit back on. This will ensure a proper system reset.)
The NSP is stuck in initialization.	NSP LCD displays Initializing for more than 2 minutes, or Central displays The System is	Poor software initialization or disconnected cables in the NSP.	Turn the NSP OFF using the rear power switch and toggle the front power switch 5 times on and off. Wait 5 minutes and then turn the NSP back on. If the NSP is still not initializing, then contact support@blackrockneuro.com

Problem	Symptom(s)	Failure	Potential fix
	Initializing.		
The instrument network has failed.	Error Message in Central and no data is transmitted or received.	Disconnected Ethernet crossover cable or the NSP is off.	Check that the NSP is powered on and that the Ethernet crossover cable is connected. Check that the Ethernet switch is powered on if it is being used.
Data recording restarts.	Error Message from Central	High data rate.	High data rates or busy hard drives can cause this error. Disable non-essential applications such as Defender, system sounds, and Automatic Updates. Ensure hard drives are not near full, defragment hard drives, or utilize hard drives with faster write speeds such as solid-state drives. Alternatively, reduce the number of continuously sampled channels.  WARNING DISABLING FIREWALLS AND ANTI-VIRUS SOFTWARE CAN LEAVE YOUR SYSTEM VULNERABLE TO CYBERATTACKS.
File Storage continuously crashes	An active recording is halted, File Storage disappears, no warnings are generated by Central.	Inadequate disk space to support high bandwidth recordings and/or File Storage is overburdening a storage device.	Do not try to record onto the same drive that the operating system is on. If the drive you are saving to is > 75% full, consider removing files from the PC's drive to long term storage. Do not delete files while you are recording, even if they are not the files that are being actively recorded. If possible, use fast writing speed drives, like SSDs.
Packets are lost	Error message from Central indicating lost data packets.	Poor Ethernet communication or excessive CPU load.	Check that the Ethernet crossover cable is Cat 6. Ensure that the Ethernet port is capable of gigabit operation. Check the Ethernet connection and reduce the CPU load. Allow Central to communicate through the firewall.

11.2 Central messages

This section describes the messages Central displays when errors or other noteworthy events occur. It also provides instructions for preventing or troubleshooting issues wherever possible.

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

Unable to allocate shared Memory.

This error message occurs when Central and another utility attempt to connect to the NSP via UDP.

To resolve this error, shut down any utilities that might be attempting to communicate with the NSP, such as cbMEX or cbSDK.

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

Recording is currently blocked by another utility. Please close that utility and try again.

This error message displays when a utility within interferes with the recording.

To resolve this error, verify impedance mode is running.

If restart the host PC and try recording again.

Unable to open the Central Application!

This error occurs when the Central Main Window startup process experiences an issue.

To resolve this error, restart the Host PC and reinstall .

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

The basis vectors have not yet been calculated for this channel. Please Build PCA then retry.

This error message displays when spike sorting settings are saved before sorting rules were generated.

Use the **Single Channel Viewer** screen to define the spike sorting rules before saving the settings.

There may not be enough data points to effectively calculate the ellipsoids. Would you like to continue?

This message displays when insufficient detected spikes have been recorded to create ellipsoid groupings on the PCA space.

To address this issue, wait until more spikes are recorded before attempting to define ellipsoids.

Note: *Recording additional spike events helps to yield accurate unit definitions. See **Spike sorting** on page 54.*

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

Unable to Open Instrument Network / It appears that the instrument network has failed.

This message displays when the system load is high.

To address this issue:

1. Confirm the following:
 - The NSP is on.
 - An Ethernet crossover cable connects the NSP and the Host PC.
 - If a switch is being used, it is connected correctly and turned on.
 - The Ethernet port has been configured as described in **10.2.2 Setting up the Ethernet connection** on page 97.
2. Restart the NSP and Host PC.

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

Network packets have been lost between the PC and the instrument.

This message displays when Central intermittently fails to communicate with the NSP over Ethernet.

To address this issue:

1. Confirm the following configurations:
 - The NSP is on and is connected to the Host PC with an Ethernet crossover cable.
2.
 - No other applications are consuming significant system resources.
 - Disable Windows Firewall, Defender, Restore, Sound, Automatic Update, System Notifications, and other Ethernet ports.
 - Uninstall anti-virus software and any unnecessary software.



WARNING

DISABLING FIREWALLS AND ANTIVIRUS SOFTWARE COULD LEAVE YOUR SYSTEM VULNERABLE TO CYBERATTACKS.

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

The system is initializing, please wait. If the system doesn't respond, click the Reset NSP button below.

This error message occurs when Central recognizes an NSP that is not initialized.

If this message displays while the NSP screen shows the **Initializing** message, wait for up to 1 minute for the error message to clear.

If the message persists after 1 minute, click the **Reset** button on the error dialog box.

If the message persists after resetting the software, the NSP.

Data Recording Restarted. The data load to the PC is extremely high and data storage has been restarted. Please reduce the burden on your PC and restart data collection.

This error occurs occasionally in very long recordings.

To address this issue:

1. Disable Windows Firewall, Defender, Restore, Sound, Automatic Update, System Notifications, and other Ethernet ports.
2. Uninstall anti-virus and any unnecessary software.
3. Close any other active applications.
4. Reduce the number and rate of continuously sampled channels.
5. Consider using a faster PC or hard drive such as a solid-state drive.

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

Library versions of Central and the NSP differ! Different Library versions are not compatible, Please check with your Blackrock representative!

This message displays when Central and the NSP use different versions of the protocol.

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

Filename.ext already exists. Overwrite?

This message displays when attempting to start a recording session in Research Interface mode and the target file name already exists.

Select **Yes** in the dialog box to start the recording session and overwrite the existing file. Select **No** to retain the existing file as is; the recording will not begin.

Electrode Configuration doesn't match Impedance File.

This message displays when loading an impedance file from the **Spike Panel > View > Show Impedance** menu containing details that do not match the correct configuration.

Check your electrode configuration and Impedance File settings to address the issue.

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

Library versions of Central and the NSP differ! Different Library versions are not compatible, Please check with your Blackrock representative!

This message displays when Central and the NSP use different versions of the protocol.

If you need additional assistance, contact Blackrock Support at support@blackrockneuro.com.

12 CLEANING, MAINTENANCE, AND DISPOSAL

12.1 Maintenance

The Cerebus Neural Processing System does not require any scheduled maintenance or calibration. Maintenance on interior components of the Cerebus system should only be performed by or on the instructions of an authorized Blackrock Microsystems representative. For more information, see **11.1 Hardware errors** on page 100 and **17 Returned merchandise authorization (RMA)** on page 112.

12.2 Disposal

Return all devices (used and unused) to a recycling point for electric and electronic devices for disposal. Do not dispose of them with household waste.

13 MAGNETIC RESONANCE

The Cerebus Neural Processing System is MR unsafe. The devices comprising the system present a projectile hazard.



WARNING

PROJECTILE HAZARD.

14 COMPLAINTS, ISSUES, AND QUESTIONS

When filing a complaint, issue, or question, provide the product description, product number, software version, lot number, complainant's name and address, and the nature of the inquiry.

Address any complaints, issues, or questions to:

<https://www.blackrockneurotech.com/support/>
support@blackrockneuro.com
U.S.: +1 801.582.5533

Notice to the user and/or patient: Any serious incident that has occurred in relation to the device should be reported to the manufacturer and the competent authority of the member state in which the user and/or patient is established.

15 WARRANTY

Blackrock Microsystems (“Blackrock”) warrants its products are free from defects in materials and manufacturing for a period of one year from the date of shipment. At its option, Blackrock will repair or replace any product that does not comply with this warranty. This warranty is voided by: (1) any modification or attempted modification to the product done by anyone other than an authorized Blackrock employee; (2) any abuse, negligent handling or misapplication of the product; or (3) any sale or other transfer of the product by the original purchaser.

Except for the warranty set forth in the preceding paragraph, Blackrock provides no warranties of any kind, either express or implied, by fact or law, and hereby disclaims all other warranties, including without limitation the implied warranties of merchantability, fitness for a particular purpose, and non-infringement of third-party patent or other intellectual property rights.

Blackrock shall not be liable for special, indirect, incidental, punitive, force majeure or consequential damages (including without limitation, damages resulting from loss of use, loss of profits, interruption, or loss of business or other economic loss) arising out of non-compliance with any warranty. Blackrock’s entire liability shall be limited to providing the remedy set forth in the previous paragraph.

16 SUPPORT

Blackrock Microsystems prides itself on its customer support. For additional information on this product or any of our products, contact our Support team through the contact information in **14 Complaints, issues, and questions** on page 109.

16.1 Downloads

Access support materials including software and application notes on the Blackrock Microsystems Support website:

<https://blackrockneurotech.com/support/>

17 RETURNED MERCHANDISE AUTHORIZATION (RMA)

In the event of a returned material authorization (RMA) or complaint, provide the product description, product number, lot number, person requesting the RMA or complaint and address, and the nature of the RMA and complaint.

In the unlikely event that your device needs to be returned to Blackrock Microsystems for repair or maintenance, do not send any equipment back without a Return Merchandise Authorization Number (RMA). An RMA number will be issued to you by a Blackrock Microsystems representative. If you need to obtain an RMA number, you may contact a product support representative at +1 801.582.5533 or by emailing support@blackrockneuro.com.

Once an RMA number has been issued, it is important to safely pack the returned item for shipping back to Blackrock Microsystems. It is preferred that you save the original boxes and packing materials that your system arrived in for return shipment. Address the package as follows:

Blackrock Microsystems, Inc.
ATTN: RMA#
630 S. Komas Dr.
Suite 200
Salt Lake City, UT 84108
USA

Tel: +1 801.582.5533